

Kingdom of Cambodia  
Nation Religion King



# EMR Clinical Data Integration Specification

V 1.1



National Payment Certification Agency  
Prepared by Department of Data Management and Information Technology

# CONTENTS

---

|                                                        |           |
|--------------------------------------------------------|-----------|
| .....                                                  | 1         |
| <b>1 OVERVIEW .....</b>                                | <b>1</b>  |
| 1.1 PURPOSE.....                                       | 1         |
| 1.2 SCOPE.....                                         | 1         |
| 1.3 INTENDED AUDIENCE .....                            | 1         |
| 1.4 IMPLEMENTATION STANDARDS.....                      | 1         |
| <b>2 INTEGRATION ARCHITECTURE &amp; DATA FLOW.....</b> | <b>1</b>  |
| 2.1 ENDPOINT SEQUENCE & DEPENDENCIES .....             | 1         |
| 2.2 HOW EMR INTEGRATION WORKS.....                     | 3         |
| 2.3 COMPONENT ARCHITECTURE.....                        | 6         |
| 2.4 DATA MODEL RELATIONSHIPS .....                     | 6         |
| 2.5 IMPLEMENTATION CONSIDERATIONS.....                 | 7         |
| 2.6 KEY FEATURES .....                                 | 7         |
| <b>3 SECURITY &amp; AUTHENTICATION.....</b>            | <b>7</b>  |
| <b>4 API ENDPOINTS .....</b>                           | <b>7</b>  |
| 4.1 AUTHENTICATION ENDPOINT.....                       | 7         |
| 4.2 SEARCH BENEFICIARY .....                           | 8         |
| 4.3 SUBMISSION ENDPOINT .....                          | 9         |
| 4.4 PROCESSING ENDPOINT .....                          | 11        |
| 4.5 TRACKING PROCESSING RESULT .....                   | 12        |
| 4.6 SUBMISSION RESPONSE HANDLING .....                 | 13        |
| <b>5 BASE COMPONENTS .....</b>                         | <b>15</b> |
| 5.1 ENCOUNTER.....                                     | 15        |
| 5.2 OBSERVATIONS.....                                  | 16        |
| <b>6 CORE COMPONENTS.....</b>                          | <b>19</b> |
| 6.1 PATIENT.....                                       | 19        |
| 6.2 VISIT.....                                         | 21        |
| 6.3 TRIAGE.....                                        | 22        |
| 6.4 VITAL SIGNS.....                                   | 23        |
| 6.5 MEDICAL HISTORIES.....                             | 25        |
| 6.6 PHYSICAL EXAMINATIONS .....                        | 27        |
| 6.7 LABORATORIES.....                                  | 29        |
| 6.8 IMAGERIES.....                                     | 35        |
| 6.9 DIAGNOSIS.....                                     | 37        |
| 6.10 SOAPS .....                                       | 38        |
| 6.11 PRESCRIPTIONS .....                               | 39        |
| 6.12 REFERRAL.....                                     | 41        |
| 6.13 CONSULTATIONS.....                                | 43        |
| 6.14 INVOICES.....                                     | 44        |
| <b>7 SPECIFIC ENCOUNTERS.....</b>                      | <b>46</b> |
| 7.1 INPATIENT ADMISSION .....                          | 46        |
| 7.2 OUTPATIENT .....                                   | 47        |
| 7.3 EMERGENCY .....                                    | 48        |
| 7.4 SURGERY .....                                      | 49        |
| 7.5 PROGRESS NOTES.....                                | 51        |

|           |                                                       |           |
|-----------|-------------------------------------------------------|-----------|
| <b>8</b>  | <b>SPECIFIC CLINICAL DATA FORM COMPONENTS .....</b>   | <b>52</b> |
| 8.1       | INTRODUCTION TO CLINICAL DATA FORM .....              | 52        |
| 8.2       | DELIVERY FORM.....                                    | 53        |
| 8.3       | OPD ANC FORM.....                                     | 54        |
| 8.4       | OPD PNC FORM.....                                     | 55        |
| 8.5       | ABORTION FORM.....                                    | 55        |
| 8.6       | BIRTH CONTROL FORM.....                               | 56        |
| 8.7       | CHILD PHYSICAL EXAMINATION (0-28 DAYS) FORM.....      | 56        |
| 8.8       | CHILD PHYSICAL EXAMINATION (1MONTH-5YEARS) FORM ..... | 57        |
| <b>9</b>  | <b>STANDARD DATA REFERENCE GLOSSARY .....</b>         | <b>58</b> |
| 9.1       | PATIENT DEMOGRAPHIC .....                             | 58        |
| 9.2       | VITAL SIGNS OBSERVATIONS.....                         | 59        |
| 9.3       | VISIT & ADMINISTRATIVE DATA .....                     | 59        |
| 9.4       | MEDICAL HISTORY COMPONENTS.....                       | 61        |
| 9.5       | PHYSICAL EXAMINATION COMPONENTS .....                 | 61        |
| 9.6       | FORM DATA.....                                        | 62        |
| <b>10</b> | <b>RELATED DOCUMENTS .....</b>                        | <b>63</b> |

# 1 OVERVIEW

---

## 1.1 PURPOSE

This document defines the technical specification for submitting clinical visit data from **external EMR systems** to the **Digital Payment Certification System**. It ensures standardized, secure, and interoperable data exchange.

## 1.2 SCOPE

This specification covers:

- **Authentication and Security:** Secure communication protocols and authentication methods
- **Data Submission:** Complete the visit data submission process
- **Data Coverage:** Clinical, administrative, and billing-related information
- **Response Handling:** Acknowledgement (ACK) and processing behavior
- **Error Management:** Validation rules and error handling procedures

## 1.3 INTENDED AUDIENCE

- **EMR Vendors:** External system developers and implementers
- **Integration Developers:** Backend and API integration specialists
- **Technical Architects:** System designers and solution architects
- **Quality Assurance Teams:** Testing and interoperability validators

## 1.4 IMPLEMENTATION STANDARDS

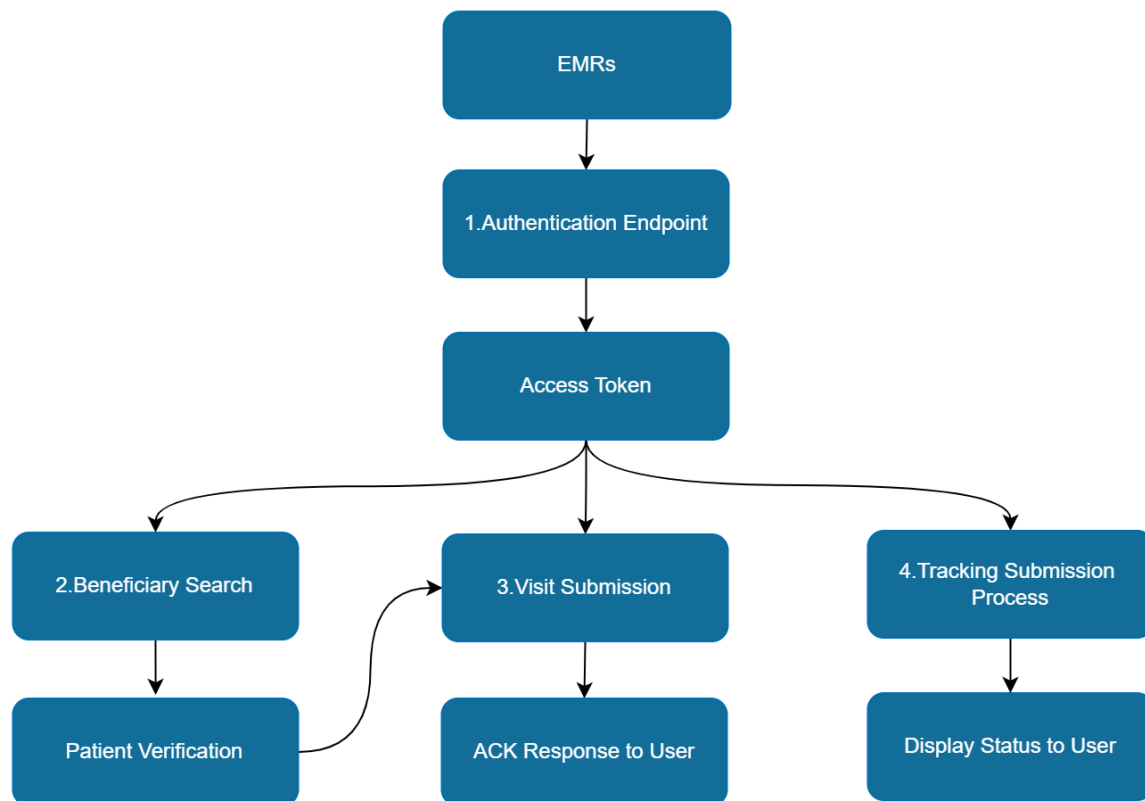
- **API Protocol:** RESTful JSON over HTTPS
- **Diagnostic Coding:** ICD-10 (Category/Subcategory codes)
- **NPCA Data Dictionary**
- **Date/Time Format:** ISO-8601 with Cambodia time zone (+07:00)
  - *Example:* 2025-11-26T11:32:00.000+07:00
- **Demographic Standards:** National Committee for Demography and Development (NCDD) standards
  - *Reference data:* <https://data.mef.gov.kh/>

# 2 INTEGRATION ARCHITECTURE & DATA FLOW

---

## 2.1 ENDPOINT SEQUENCE & DEPENDENCIES

The integration involves three primary API endpoints that work together to manage patient visits:



### Endpoint 1: Authentication

- **Purpose:** Obtain a secure OAuth 2.0 access token (Bearer token) for authenticating all subsequent API calls.
- **When to Use:** Before any API call; upon token expiration (via 401 Unauthorized response); or as part of a scheduled refresh cycle.
- **Frequency:** Initial setup and periodic refresh (tokens are typically valid for 2 hours).
- **Dependencies:** None. This is the entry point.

### Endpoint 2: Beneficiary Search

- **Purpose:** Verify if the card exists, has a benefit program, and has basic demographic information.
- **When to Use:** **Strongly recommended** before visit submission to ensure data linkage and avoid errors.
- **Frequency:** Once per patient/visit verification.
- **Dependencies:** Requires a valid **access token** from Endpoint 1.
- **Output:** Critical Eligibility card, SPID used in the Visit Submission payload.

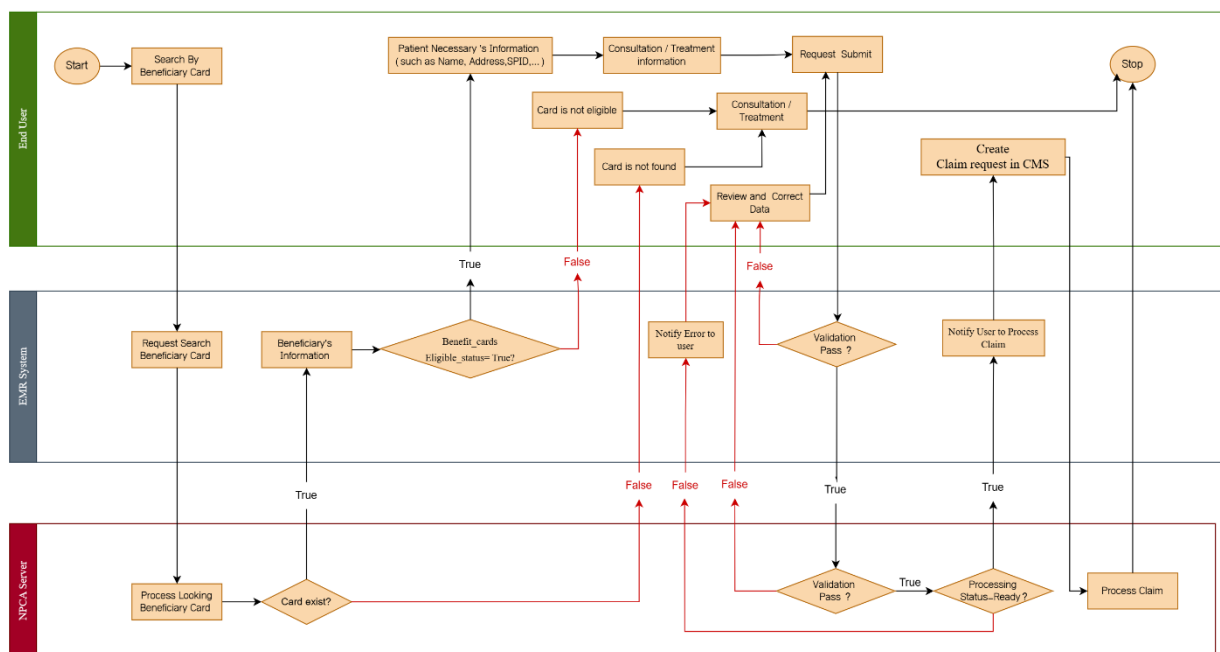
### Endpoint 3: Visit Submission

- **Purpose:** Submit complete or partial clinical visit data for processing and storage.
- **When to Use:** After successful patient verification (Endpoint 2) and with a valid access token (Endpoint 1).
- **Frequency:** Once per patient visit event (create/update).
- **Dependencies:** Requires a valid **access token** (Endpoint 1) and a SPID (obtained via Endpoint 2).
- **Response:** Immediate HTTP 200 Accepted with a **status**, **visit\_ulid** and **visit\_code**.

## Endpoint 4: Tracking Processing Result

- **Purpose:** Retrieve the latest processing status and validation result of a previously submitted visit, including whether it has been accepted, rejected, is still processing, or requires user action.
- **When to Use:** After a visit has been submitted to the Integration Platform (Endpoint 3) and with a valid access token (Endpoint 1). The EMR needs to check the current processing status or obtain the final processing result. This is especially useful when processing is asynchronous or when the initial submission response only acknowledges receipt.
- **Frequency:** On demand or periodically until processing is completed. Recommended polling interval: every 8 hours (or according to system guidelines). Stop polling once the status becomes **Completed**, **Rejected**, or **Failed**.
- **Dependencies:** Authorized API credentials and Visit Submission
- **Response:** Returns the **status** (**Processing**, **Pending**, **Failed**, **Ready**), processed timestamps, Validation or processing failure reason, processing status, and any identifiers such as Visit Code.

## 2.2 HOW EMR INTEGRATION WORKS



*Flow of EMR Integration*

### Step 1: Beneficiary Lookup

This is where the process starts. User swipes or enters the patient's card number into the system.

- **Action:** User swipes or manually enters the patient's benefit card number.
- **Outcomes:**
  1. If the card is **found**, the system retrieves the **Patient's Details** (Home, Address, SPID, etc.). The "Patient's Information" is prepared, and the process moves to step 2.
  2. If the **Card is not found** in the database, the patient would **not be eligible to receive treatment under the benefit**.

### Step 2: Eligibility & Card Validity Check

- **Action:** System verifies eligibility status linked to the patient's record.
- **Outcomes:**
  1. **Eligible:** Patient information is confirmed and prepared. The patient may proceed to receive covered treatment.
  2. **Not Eligible:** Benefit does **not apply**; treatment will not be covered.

### Step 3: Benefit Request Submission

#### 1. Submission Initiation

- OPD/IPD: Submit after treatment completion
- IPD-HEF-Poor: Submit after admission (pre-authorization/coverage guarantee)

#### 2. EMR Portal Validation

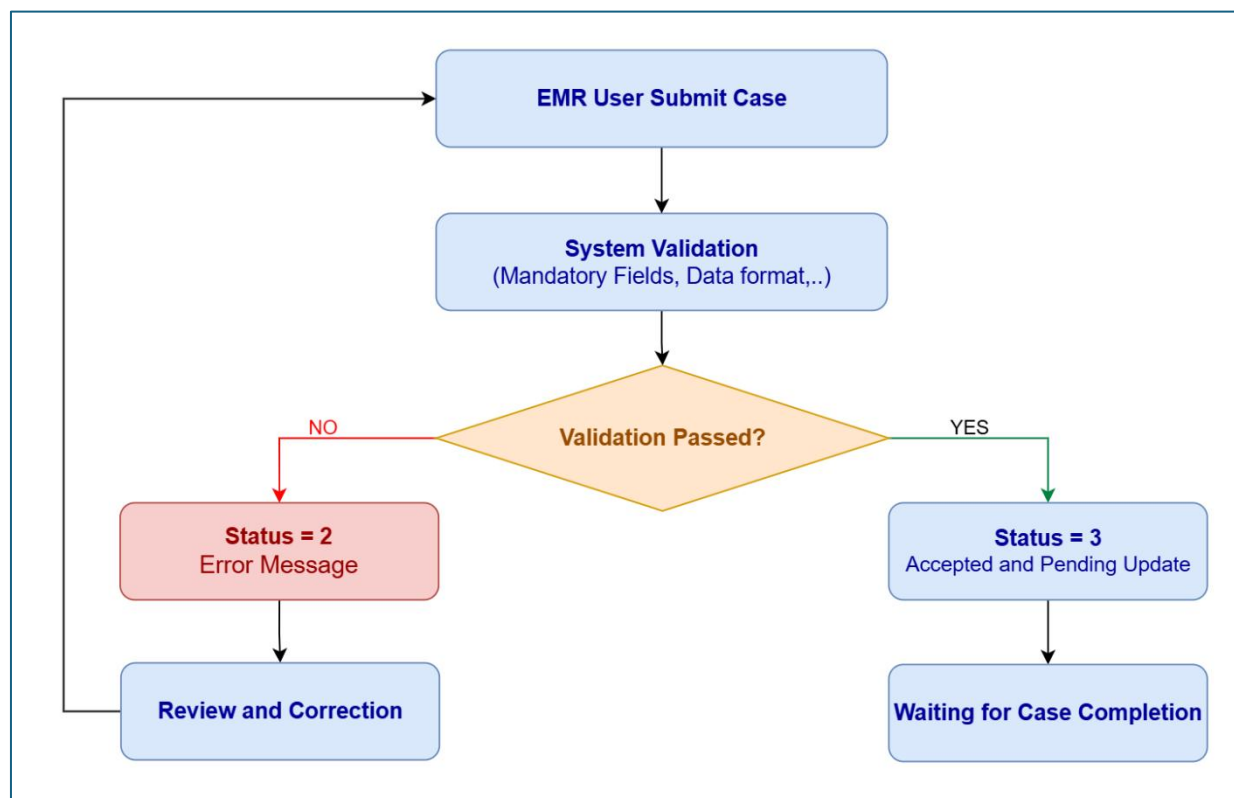
- **Process:** Upon submission, the EMR portal performs local validation and verifies required fields (diagnosis, procedures, etc.).
- **Outcomes:**
  - **Passed:** Request is queued for **NPCA server validation**.
  - **Failed:** Returns **error message** to user with details (e.g., "Missing diagnosis code"). User must correct and resubmit.

#### 3. NPCA Server Validation

After EMR validation passes, the request is sent to the **Digital Payment Certification System**.

- Validates patient eligibility status.
- Performs policy matching (procedure coverage, etc.).
- Generates a unique reference number(ulid) if processed successfully.

### 2.2.1 Incomplete Case Workflow

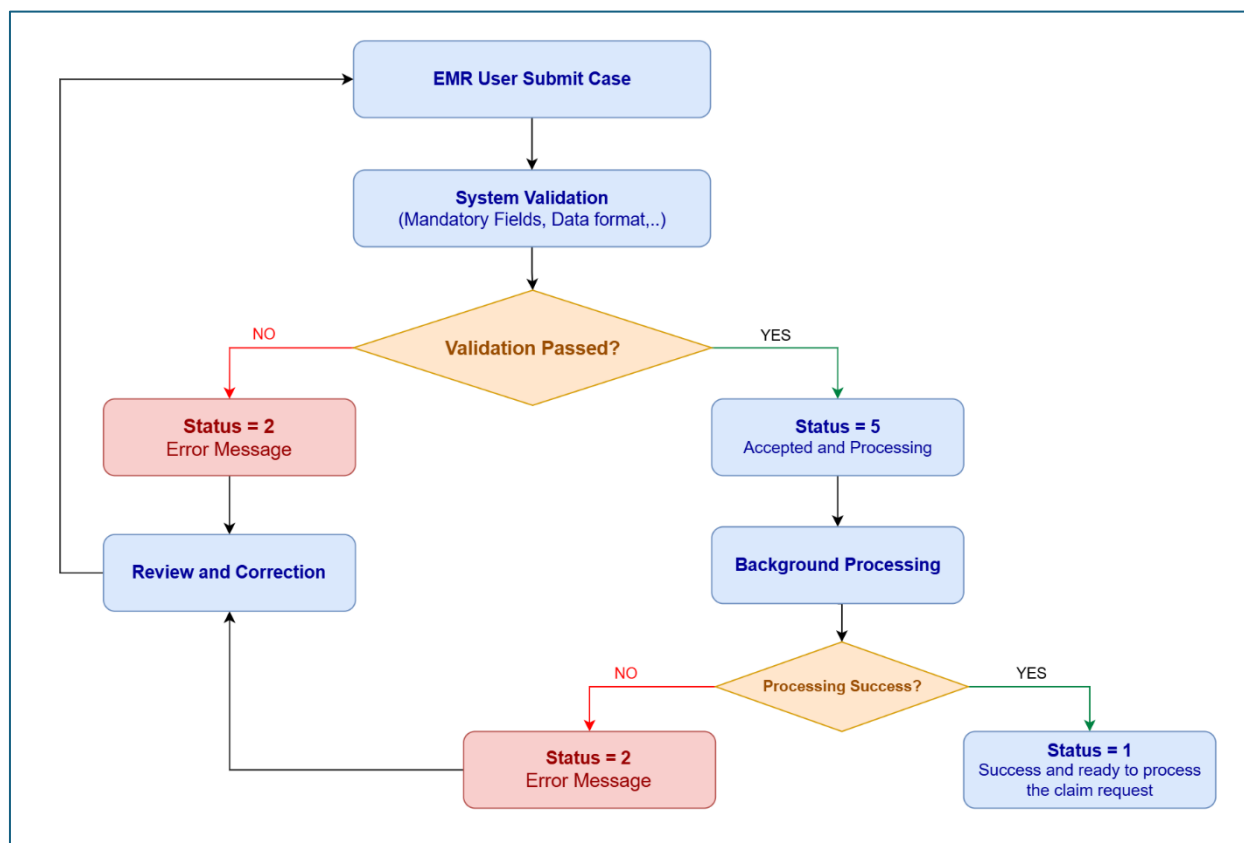


*Flow of Incomplete Case*

An incomplete case is a visit for which patient care is still in progress. As the treatment has not yet been completed, the visit record remains open and may be updated until the case is finalized.

| Status | Meaning                                                                                                                                                                                                                                                                                                                                                              | User Action                                                                                                                                                                                                                                                                                                                                 |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2      | Submission failed                                                                                                                                                                                                                                                                                                                                                    | Correct the data and resubmit                                                                                                                                                                                                                                                                                                               |
| 3      | <ul style="list-style-type: none"><li>The visit has been accepted by the Integration Platform.</li><li>The case is not yet complete and requires further updates.</li><li>For eligible HEF IPD patients, users may process the initial claim, such as food allowance.</li><li>Claim requests cannot be submitted until the visit is completed and updated.</li></ul> | <p>After the patient completes treatment, the healthcare provider must update the visit with the final information and resubmit it to complete the processing workflow.</p> <p><b>Note:</b> <i>Eligible for HEF food allowance → Healthcare provider submits daily food allowance claims through the Claim Management System (CMS).</i></p> |

## 2.2.2 Complete Case Workflow



*Flow of Complete Case*

A complete case refers to a patient visit that has been completed. The patient has finished treatment, and the visit record contains the final information required for processing.

| Status | Meaning                           | User Action                        |
|--------|-----------------------------------|------------------------------------|
| 5      | Accepted and processing           | Wait for the final result          |
| 1      | Processing completed successfully | The claim request can be initiated |
| 2      | Submission failed                 | Correct the data and resubmit      |

## 2.3 COMPONENT ARCHITECTURE

The API follows a modular architecture with interconnected components that represent different aspects of healthcare delivery:

1. **Base Components** - Learn about fundamental structures like Encounters and Observations that serve as building blocks for more complex operations
2. **Core Components** - Essential models such as Patient, Visit, Triage, Treatment, Invoices, and other Clinical that form the core of the healthcare information system
3. **Specific Encounters** - Supporting structures categorized encounters that can easily help identify each case of a patient visit

## 2.4 DATA MODEL RELATIONSHIPS

The components are interconnected through reference codes:

- Each Patient can have multiple Visits
- Each Visit can include multiple Encounters (Outpatient, Inpatient, Emergency)
- Each Encounter can have associated clinical data (Observations, Diagnoses, Prescriptions, etc.)

## 2.5 IMPLEMENTATION CONSIDERATIONS

- Designed for interoperability with healthcare standards
- Supports Cambodia-specific requirements (local administrative divisions, facility structures)
- Flexible observation model accommodates various data types

## 2.6 KEY FEATURES

- Flexible and extensible components
- Comprehensive medical history tracking
- Support for different encounter types (outpatient, inpatient, emergency)
- Integrated billing and payment systems
- Referral management between healthcare facilities
- Detailed clinical documentation with SOAP notes

## 3 SECURITY & AUTHENTICATION

| Parameter        | Specification                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protocol         | HTTPS (TLS 1.2 or higher)                                                                                                                                                                           |
| Authentication   | OAuth 2.0 Bearer Token                                                                                                                                                                              |
| Required Headers | Authorization: Bearer <access_token><br>Content-Type: application/json<br>Accept: application/json<br>X-System-Code: <i>[provided by NPCA]</i><br>X-Health-Facility-Code: <i>[provided by NPCA]</i> |

## 4 API ENDPOINTS

This API provides endpoints to manage patient visits in the healthcare system. It allows submitting patient visits along with related data such as triage information, encounters, diagnoses, prescriptions, and more.

### 4.1 AUTHENTICATION ENDPOINT

The API uses OAuth 2.0 with Password Client for authentication.

|        |              |
|--------|--------------|
| URL    | /oauth/token |
| Method | POST         |

|              |                                                                                                                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Request Body | <p>Object:</p> <ul style="list-style-type: none"> <li>• grant_type: String</li> <li>• client_id: String</li> <li>• client_secret: String</li> <li>• username: String</li> <li>• password: String</li> </ul> <p><i>NPCA Team will provide credential information.</i></p> |
| Response     | <p>Object:</p> <ul style="list-style-type: none"> <li>• token_type: String</li> <li>• access_token: String (base 64)</li> <li>• expires_in: Timestamp</li> <li>• refresh_token: String (base 64)</li> </ul>                                                              |

## 4.2 SEARCH BENEFICIARY

|          |                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL      | api/v1/external/beneficiary/search                                                                                                                                                                                                                                                                                                                                                   |
| Method   | GET                                                                                                                                                                                                                                                                                                                                                                                  |
| Params   | card_number                                                                                                                                                                                                                                                                                                                                                                          |
| Response | <p>Objects (can provide beneficiary people):</p> <ul style="list-style-type: none"> <li>• spid: String</li> <li>• patient_code: String</li> <li>• surname: String</li> <li>• name: String</li> <li>• sex: String</li> <li>• birthdate: YYYY-mm-dd</li> <li>• phone_number: String</li> <li>• nationality: String</li> <li>• relationship_to_head</li> <li>• dead: Boolean</li> </ul> |

|  |                                                                                                                                                                                                                                                                                                    |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• death_date: datetime</li> <li>• created_at: datetime</li> <li>• updated_at: datetime</li> <li>• address: object</li> <li>• benefit_cards: array</li> <li>• identifications: array</li> <li>• occupation: string</li> <li>• disabilities: array</li> </ul> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4.3 SUBMISSION ENDPOINT

|              |                                                                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL          | /api/external/v1/visits                                                                                                                                                                                                  |
| Method       | POST                                                                                                                                                                                                                     |
| Request Body | <i>Depending on the case and actual treatment procedure, refer to section 4.3.1 and Sample Json[1]</i>                                                                                                                   |
| Response     | <pre>{   "success": true,   "http_code": 200,   "message": <i>message</i>,   "data": {     "status": <i>status_code</i>,     "uuld": "<i>unique code generated</i>",     "visit_code": "<i>unique code</i>",   } }</pre> |

### 4.3.1 Request Body Structure

The following is an overview of the JSON data structured for submission to the endpoint. Each is a component that will be defined later in the document.

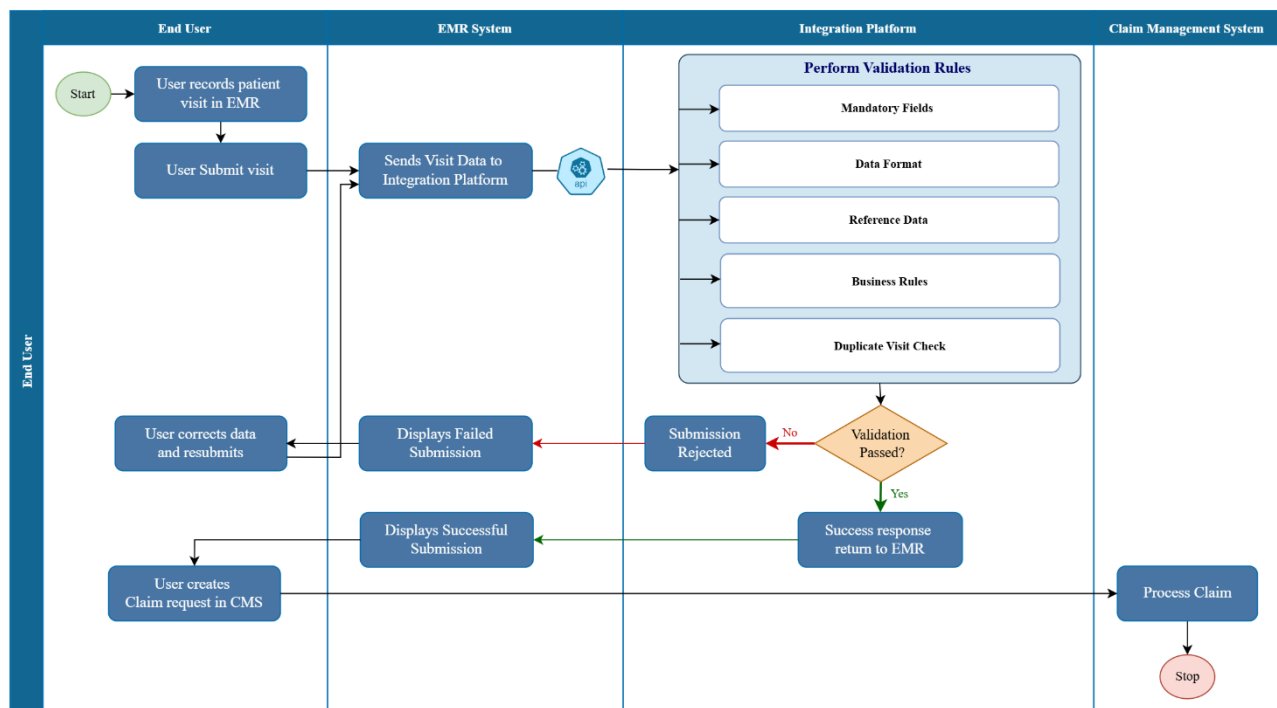
```
{
  "visits": [{
    "patient": {},
```

```

"triages": [{}],
"outpatients": [{}],
"inpatients": [{}],
"emergencies": [{}],
"surgeries": [{}],
"progress_notes ": [{}],
"soaps ": [{}],
"vital_signs": [{}],
"medical_histories": [{}],
"physical_examinations": [{}],
"laboratories": [{}],
"imageries": [{}],
"diagnoses": [{}],
"prescriptions": [{}],
"invoices": [{}],
"referral": {}
}
}
}

```

### 4.3.2 Validation Rules



*Flow of Visit Submission Validation and Claim Process*

| N0 | Category                    | Description                                                                                         |
|----|-----------------------------|-----------------------------------------------------------------------------------------------------|
| 1  | Mandatory Fields Validation | The system validates that all required information is provided before accepting a visit submission. |
| 2  | Data Format                 | The system validates all submitted data formats.                                                    |
| 3  | Reference data              | The system validates submitted codes against approved reference data.                               |
| 4  | Business rules              | The system validates operational and clinical business rules.                                       |

|   |                       |                                                                                                                  |
|---|-----------------------|------------------------------------------------------------------------------------------------------------------|
| 5 | Duplicate Visit check | The system checks for duplicate submissions. The same visit must not be submitted multiple times as a new visit. |
|---|-----------------------|------------------------------------------------------------------------------------------------------------------|

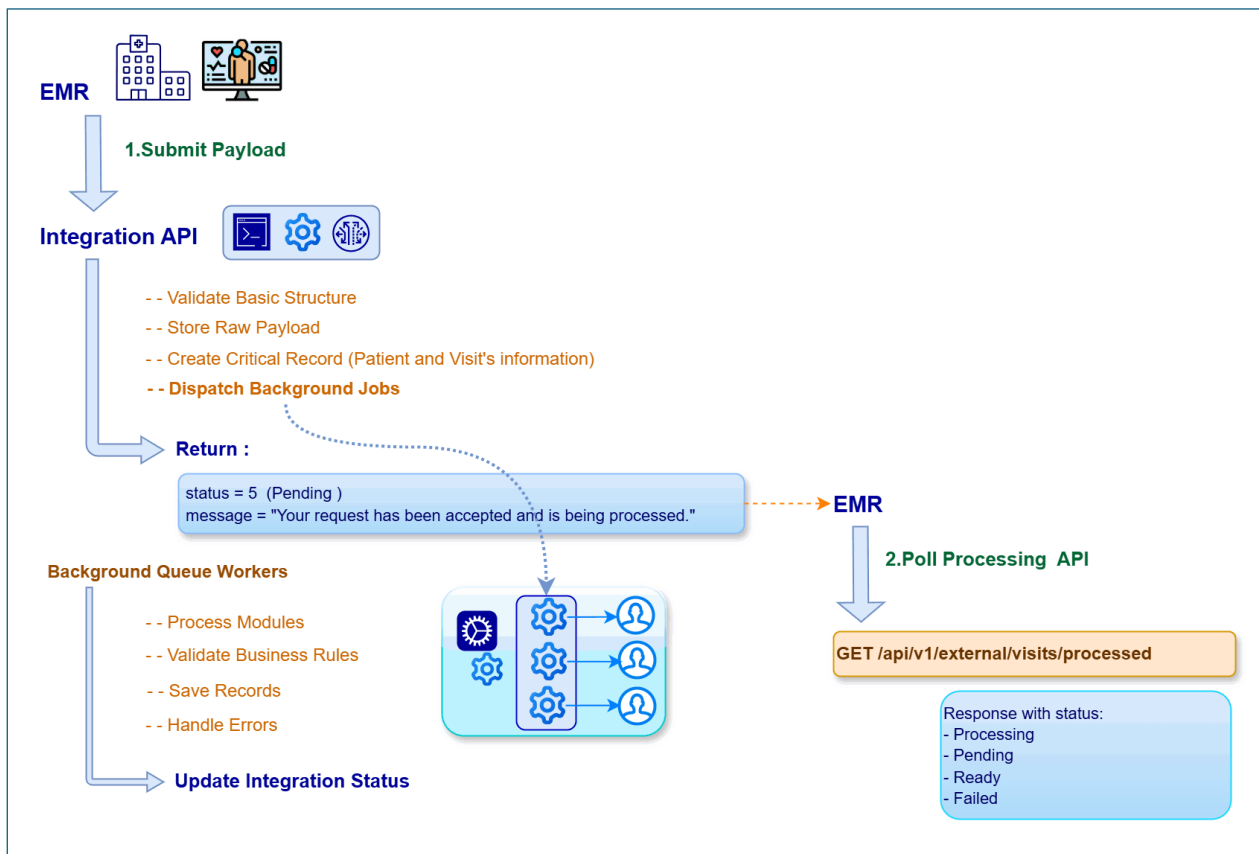
| N0 | Business Rules Validation       | Description                                                                                                                                                                                                |
|----|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Access & Patient Identity Rules | Verify the submitting facility is authorized, and the patient identity is valid and eligible.                                                                                                              |
| 2  | Basic Visit Rules               | Validate visit type, admission and discharge dates, visit status, unique visit code, and other core visit information.                                                                                     |
| 3  | Completed Case Rules            | Ensure all required clinical and financial information has been submitted, including triage, diagnoses, procedures, invoices, and OPD/IPD-specific data.                                                   |
| 4  | Clinical Data Linking Rules     | Verify relationships between encounters and associated clinical records such as vital signs, SOAP notes, diagnoses, prescriptions, laboratory orders, imaging, physical examinations, and medical history. |
| 5  | Resubmission Rules              | Validate whether the visit is eligible for resubmission based on its current processing status, admission changes, locked records, pending validation, or previous submission results.                     |

## 4.4 PROCESSING ENDPOINT

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URL      | /api/v1/external/visits/processed                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Params   | <ul style="list-style-type: none"> <li>discharged_at_from: YYYY-mm-dd</li> <li>discharged_at_to: YYYY-mm-dd</li> <li>admitted_at_from: YYYY-mm-dd</li> <li>admitted_at_to: YYYY-mm-dd</li> <li>status: processing / failed / ready / pending</li> <li>visit_code: String</li> <li>per_page: Number (Number of items per page)</li> </ul> <p><i>Importance: Apply this enhancement to improve the identification of records that require updates.</i></p> |
| Response | <p>Object</p> <ul style="list-style-type: none"> <li>success: Boolean</li> <li>http_code: Number</li> <li>message: String</li> <li>data: Objects</li> </ul>                                                                                                                                                                                                                                                                                              |

## 4.5 TRACKING PROCESSING RESULT

### 4.5.1 Submission API Flow



*Flow of Submission Tracking Process*

### 4.5.2 Implement Processing Status Retrieval

EMR systems must retrieve processing results after a visit enters Status 5. This ensures users receive final processing outcomes.

Benefits:

- Automatic update on EMR UI
- No manual checking
- Better user experience

### 4.5.3 Display Processing Status

Users should be able to see:

- Processing
- Failed
- Ready
- Pending

directly from the EMR interface.

#### 4.5.4 Submission Status Reference

| Status | Processing Status | Meaning                                                         |
|--------|-------------------|-----------------------------------------------------------------|
| 5      | Processing        | Background processing in progress                               |
| 3      | Pending           | Additional updates are required before processing can continue. |
| 2      | Failed            | Validation or processing failure                                |
| 1      | Ready             | Processing completed successfully                               |

#### 4.5.5 Display Failure Reasons

When processing fails, the EMR system should display the reason for the failure returned by the Integration Platform.

Examples:

- Referral reason is required.
- Diagnosis code is invalid.
- Visit Code has been submitted.
- Systolic Blood Pressure must be a number.

#### 4.5.6 Support Data Correction and Resubmission

When processing fails:

1. User reviews the failure reason.
2. User corrects the data.
3. User resubmits the visit.

The EMR system must support this workflow.

### 4.6 SUBMISSION RESPONSE HANDLING

#### 4.6.1 Success Responses

| Response                                                                                                                                                                                                                                                 | Description                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <pre>{<br/>  "success": true,<br/>  "http_code": 200,<br/>  "message": "Your request is accepted and pending update.",<br/>  "data": {<br/>    "visit_ulid": "{ ulid }",<br/>    "visit_code": "{ visit_code }",<br/>    "status": 3<br/>  }<br/>}</pre> | Submission has been accepted, and additional updates are required before processing can continue. |
| <pre>{<br/>  "success": true,</pre>                                                                                                                                                                                                                      | Submission has been accepted and is being processed in the background.                            |

|                                                                                                                                                                                       |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre> "http_code": 200, "message": "Your request is accepted and is being processed.", "data": {   "visit_ulid": "{ ulid }",   "visit_code": "{ visit_code }",   "status": 5 } </pre> |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

## 4.6.2 Error Response

### 4.6.2.1 400 Bad Request

```

{
  "success": false,
  "http_code": 400,
  "message": "Invalid request data",
  "errors": "Error Message"
}

```

### 4.6.2.2 401 Unauthorized

```

{
  "success": false,
  "http_code": 401,
  "message": "Unauthenticated",
  "errors": "Invalid or expired token"
}

```

### 4.6.2.3 403 Forbidden

```

{
  "success": false,
  "http_code": 403,
  "message": "Access denied",
  "errors": "You do not have permission to create visits"
}

```

### 4.6.2.4 500 Server Error

```

{
  "success": false,
  "http_code": 403,
  "message": "Internal server error",
  "errors": "An unexpected error occurred while processing your request"
}

```

### 4.6.2.5 Validation Error

| Format | Example |
|--------|---------|
| {      | {       |

|                                                                                                                                                        |                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> "success": false, "http_code": 422, "message": "Invalid request data", "errors": "{ error_field_name } { error_message }", "data": null } </pre> | <pre> "success": false, "http_code": 422, "message": "Invalid request data", "errors": "The visits.0.code field is required.", "data": null } </pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

## 5 BASE COMPONENTS

This section attempts to explain the basic concept of each component used under the visit component.

### 5.1 ENCOUNTER

An Encounter represents a clinical interaction between a patient and a healthcare provider during a visit.

#### 5.1.1 Data Model

| Field          | Type     | Required | Description                                                 |
|----------------|----------|----------|-------------------------------------------------------------|
| patient_code   | string   | YES      | Reference to the patient's unique code                      |
| visit_code     | string   | YES      | Reference to visit the unique code                          |
| code           | string   | YES      | Unique identifier for the encounter                         |
| class          | string   | YES      | Class of encounter ("IPD", "OPD", or "EMERGENCY")           |
| name           | string   | YES      | What is the encounter (triage, ward names, sub-clinic ... ) |
| service_type   | string   | YES      | Type of the service provided (nullable)                     |
| started_at     | datetime | YES      | Date and time encounter started                             |
| ended_at       | datetime | YES      | Date and time encounter ended (nullable)                    |
| encountered_by | string   | YES      | Name of healthcare provider                                 |
| title          | string   | YES      | Whether the healthcare provider is doctor or nurse          |
| created_at     | datetime | YES      | Record creation timestamp                                   |
| updated_at     | datetime |          | Record the last update timestamp                            |

#### 5.1.2 Example

```

{
  "patient_code": "P25003877",
  "visit_code": "V420650",
  "code": "E317001",
  "class_name": "OPD",
  "name": "General Medicine"
  "service_type": "General Consultation",
  "started_at": "2025-03-17T09:35:00.000+07:00",
  "ended_at": "2025-03-17T10:15:00.000+07:00",
  "encountered_by": "Dr. Chhun Sovannarith"
  "title": "Doctor",

```

```

"created_at": "2025-03-17T09:35:00.000+07:00",
"updated_at": "2025-03-17T10:15:00.000+07:00"
}

```

## 5.2 OBSERVATIONS

Observations contain clinical data about a patient. They are designed to be flexible enough to capture virtually any type of patient information.

### 5.2.1 Data Model

| Field            | Type    | Required | Description                                                                                             |
|------------------|---------|----------|---------------------------------------------------------------------------------------------------------|
| encounter_code   | string  | YES      | Reference to encounter unique code                                                                      |
| observation_code | string  |          | Reference to parent observation (nullable)                                                              |
| name             | string  | YES      | Name for the observation                                                                                |
| category         | string  | YES      | Name for the category of the observation if applicable                                                  |
| value            | various | YES      | Value of the observation (type depends on value_type)                                                   |
| value_type       | string  | YES      | Type of value ("boolean", "string", "text", "integer", "float", "datetime", "time", "array", "complex") |
| value_unit       | string  |          | Unit of measurement (nullable)                                                                          |

### 5.2.2 Value Types

The observation structure needs to be flexible so any type of data stored in the EMR can be submitted via API. To introduce flexibility, a value type is necessary. It basically determines what type of data it is in the value field.

#### 5.2.2.1 Boolean Example

```

{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Smoker",
  "category": "Social History",
  "value": false,
  "value_type": "boolean",
  "value_unit": null
}

```

#### 5.2.2.2 String Example

```

{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Blood Type",
  "category": "Laboratory",
  "value": "O+",
  "value_type": "string",
  "value_unit": null
}

```

#### 5.2.2.3 Text Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Clinical Notes",
  "category": "Assessment",
  "value": "Patient reports symptoms of malaria including periodic
fever, headache, and fatigue. Has history of travel to Mondulkiri
province two weeks ago.",
  "value_type": "text",
  "value_unit": null
}
```

#### 5.2.2.4 Numeric Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Blood Glucose",
  "category": "Vital Signs",
  "value": 110,
  "value_type": "integer",
  "value_unit": "mg/dL"
}
```

#### 5.2.2.5 Datetime Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Last Meal Time",
  "category": "Nutrition",
  "value": "2025-03-17T07:00:00.000+07:00",
  "value_type": "datetime",
  "value_unit": null
}
```

#### 5.2.2.6 Time Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Duration of Symptoms",
  "category": "History",
  "value": "48:00:00",
  "value_type": "time",
  "value_unit": "hh:mm:ss"
}
```

### 5.2.2.7 Array Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Symptoms",
  "category": "History",
  "value": ["Fever", "Headache", "Fatigue", "Joint Pain"],
  "value_type": "array",
  "value_unit": null
}
```

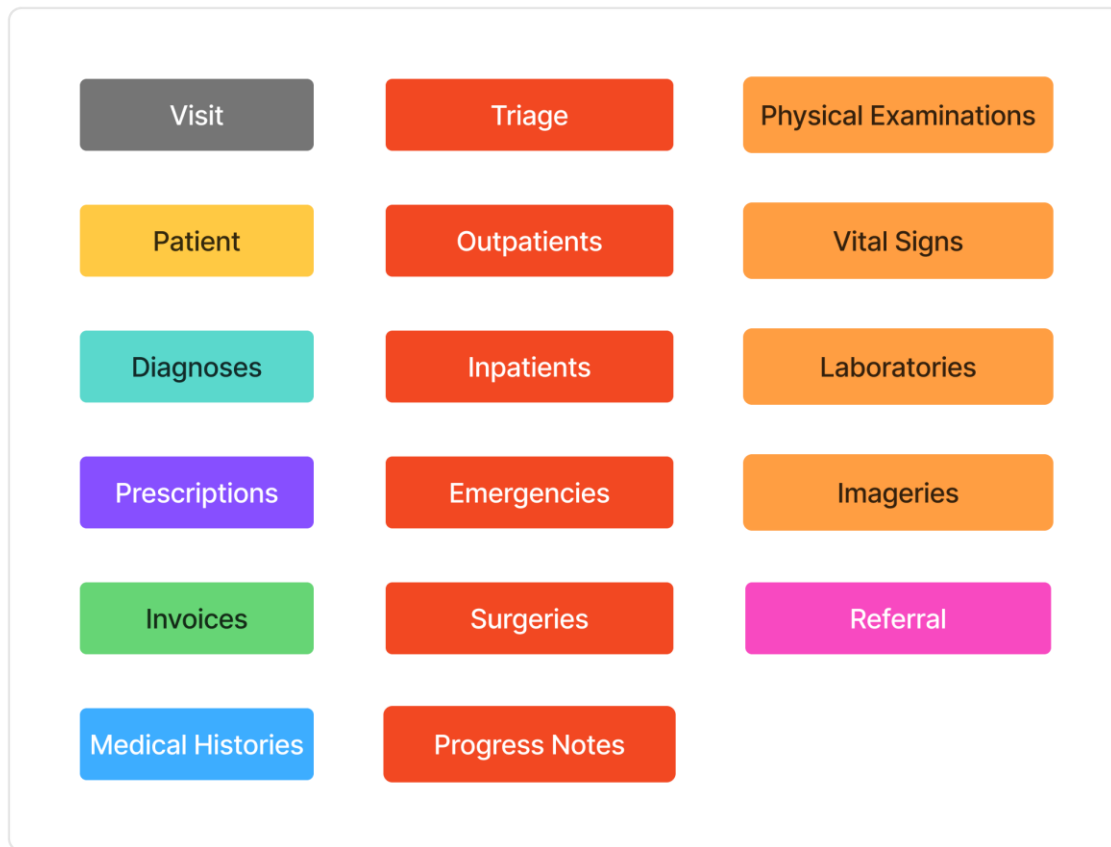
### 5.2.2.8 Complex Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Malaria Risk Assessment",
  "category": "Risk Assessment",
  "value": {
    "risk_level": "High",
    "travel_history": true,
    "prevention_measures": ["Bed nets", "Repellent"],
    "endemic_area": "Mondulkiri province"
  },
  "value_type": "complex",
  "value_unit": null
}
```

**Note:** conditions also have the same structure and values as observations. But for simplicity, the conditions in the Medical History components are shortened and unified to make it easier to implement them.

## 6 CORE COMPONENTS

---



### 6.1 PATIENT

The Patient object contains personal information and demographics for an individual under medical care.

#### 6.1.1 Data Model

| Field          | Type             | Required | Description                                     |
|----------------|------------------|----------|-------------------------------------------------|
| code           | string           | YES      | Unique identifier for the patient in the system |
| surname        | string           | YES      | Patient's family name                           |
| name           | string           | YES      | Patient's given name                            |
| sex            | string           | YES      | Patient's biological sex                        |
| birthdate      | date             | YES      | Patient's date of birth                         |
| phone          | string           | YES      | Patient's contact number                        |
| nationality    | string           | YES      | Patient's nationality                           |
| disabilities   | array of strings |          | List of disabilities, if applicable             |
| occupation     | string           |          | Patient's job or profession (nullable)          |
| marital_status | string           | YES      | Patient's marital status (nullable)             |
| photos         | array of strings |          | URLs to patient photos (nullable)               |

|                 |                  |     |                                                             |
|-----------------|------------------|-----|-------------------------------------------------------------|
| address         | object           | YES | Patient's address information (see Address model)           |
| identifications | array of objects |     | Patient identification documents (see Identification model) |
| death_date      | datetime         |     | Date and time of death, if applicable                       |
| spid            | string           | YES | Patient's Social Protection ID                              |
| created_at      | datetime         | YES | Record creation timestamp                                   |
| updated_at      | datetime         | YES | Record the last update timestamp                            |

### 6.1.2 Address Model

| Field         | Type   | Required | Description                            |
|---------------|--------|----------|----------------------------------------|
| province      | object | YES      | Information about the province         |
| province.code | string | YES      | Province code from NCDD standard       |
| province.name | string | YES      | Province's name in Khmer               |
| district      | object | YES      | Information about the district         |
| district.code | string | YES      | District code from NCDD standard       |
| district.name | string | YES      | District name in Khmer                 |
| commune       | object | YES      | Information about the commune          |
| commune.code  | string | YES      | Commune code from NCDD standard        |
| commune.name  | string | YES      | Commune name in Khmer                  |
| village       | object | YES      | Information about the village          |
| village.code  | string | YES      | Village code from NCDD standard        |
| village.name  | string | YES      | Village name in Khmer                  |
| house_number  | string |          | House number (nullable)                |
| street_number | string |          | Street number (nullable)               |
| location      | string |          | Additional location details (nullable) |

### 6.1.3 Identification Model

| Field        | Type   | Required | Description                      |
|--------------|--------|----------|----------------------------------|
| patient_code | string | YES      | Reference to patient unique code |
| card_code    | string | YES      | Identification number            |
| card_type    | string | YES      | Type of identification document  |

### 6.1.4 Example

```
{
  "patient": {
    "code": "P25003877",
    "surname": "សៅ",
    "name": "សុភ័ន្ទ",
    "sex": "M",
```

```

"birthdate": "1980-05-15",
"phone": "012 345 678",
"nationality": "Cambodian",
"disabilities": ["Hearing impairment"],
"occupation": "Teacher",
"marital_status": "Married",
"photos": ["https://hospital.example/photos/PT20250317001_1.jpg"],
"address": {
  "province": {
    "code": "12",
    "name": "Phnom Penh"
  },
  "district": {
    "code": "1201",
    "name": "Chamkar Mon"
  },
  "commune": {
    "code": "120101",
    "name": "Tonle Bassac"
  },
  "village": {
    "code": "12010101",
    "name": "Phsar Deum Thkov"
  },
  "house_number": "45",
  "street_number": "302",
  "location": "Near Central Market"
},
"identifications": [
  {
    "patient_code": "P25003877",
    "card_code": "ID123456789",
    "card_type": "National ID"
  }
],
"death_date": null,
"spid": "250317001",
"created_at": "2025-03-17T08:30:00.000+07:00",
"updated_at": "2025-03-17T08:30:00.000+07:00"
}
}

```

## 6.2 VISIT

A Visit represents a patient's stay at a healthcare facility, from admission to discharge.

### 6.2.1 Data Model

| Field                | Type   | Required | Description                                                   |
|----------------------|--------|----------|---------------------------------------------------------------|
| health_facility_code | string | YES      | Unique identifier for the healthcare facility, issued by NPCA |

|                |          |     |                                                            |
|----------------|----------|-----|------------------------------------------------------------|
| patient_code   | string   | YES | Reference to the patient's unique code                     |
| code           | string   | YES | Unique identifier for the visit                            |
| admission_type | string   | YES | How a patient is admitted to a healthcare facility         |
| discharge_type | string   | YES | How a patient is discharged from a healthcare facility     |
| visit_outcome  | string   | YES | The patient's condition at the end of the visit            |
| visit_type     | string   | YES | The type or purpose of a patient's visit (IPD or OPD)      |
| admitted_at    | datetime | YES | Date and time of admission                                 |
| discharged_at  | datetime | YES | Date and time of discharge                                 |
| followup_at    | datetime |     | When the patient is scheduled to return for follow-up care |
| created_at     | datetime | YES | Record creation timestamp                                  |
| updated_at     | datetime | YES | Record the last update timestamp                           |

## 6.2.2 Example

```
{
  "visits": [
    {
      "health_facility_code": "121020",
      "patient_code": "P25003877",
      "code": "V420650",
      "admission_type": "Walk-in",
      "discharge_type": "Authorized",
      "visit_outcome": "Improved",
      "visit_type": "OPD",
      "admitted_at": "2025-03-17T09:30:00.000+07:00",
      "discharged_at": "2025-03-17T10:45:00.000+07:00",
      "followup_at": "2025-03-24T09:00:00.000+07:00",
      "created_at": "2025-03-17T09:30:00.000+07:00",
      "updated_at": "2025-03-17T10:45:00.000+07:00"
    }
    ...
  ]
}
```

## 6.3 TRIAGE

Triage represents the initial assessment of a patient to determine treatment priority.

### 6.3.1 Data Model

| Field          | Type   | Required | Description                                     |
|----------------|--------|----------|-------------------------------------------------|
| patient_code   | string | YES      | Reference to patient's unique code              |
| visit_code     | string | YES      | Reference to visit the unique code              |
| encounter_code | string | YES      | Reference to encounter a unique code (nullable) |
| code           | string | YES      | Unique identifier for the triage                |

|                 |          |     |                                                                             |
|-----------------|----------|-----|-----------------------------------------------------------------------------|
| chief_complaint | text     | YES | Primary symptom or problem reported by the patient at the time of the visit |
| height          | number   | YES | Value of height of the patient in CM                                        |
| weight          | number   | YES | Value of weight of the patient in KG                                        |
| recorded_at     | datetime | YES | Date and time triage was recorded                                           |
| recorded_by     | string   | YES | Name of the healthcare provider who recorded the triage                     |
| title           | string   |     | Position of the healthcare provider                                         |

### 6.3.2 Example

```
{
  "triages": [{
    "visit_code": " V420650",
    "encounter_code": "P25003877",
    "code": "TR250317",
    "chief_complaint": "Periodic fever with chills for 3 days",
    "height": 170,
    "weight": 65,
    "recorded_at": "2025-03-17T09:32:00.000+07:00",
    "recorded_by": "Khun Srey Mom",
    "title": "Doctor",
  }]
}
```

## 6.4 VITAL SIGNS

Vital signs contain key physiological measurements of a patient. This object must contain all the vital signs information, such as temperature, pulse (heart rate), respiratory rate, systolic blood pressure, diastolic blood pressure, oxygen saturation, and an optional random rapid blood glucose test.

| Vital Sign                            | Normal Range                                 | Units |
|---------------------------------------|----------------------------------------------|-------|
| Temperature                           | 36.5–37.3 °C                                 | °C    |
| Pulse (Heart Rate)                    | 60–100 beats per minute                      | /mn   |
| Respiratory Rate                      | 12–20 breaths per minute                     | /mn   |
| Blood Pressure                        | Systolic: 90–120 mmHg, Diastolic: 60–80 mmHg | mmHg  |
| Oxygen Saturation (SpO <sub>2</sub> ) | 95–100%                                      | %     |
| Random Glucose Rapid Test             | Random: < 140 mg/dL, Fasting: 70–99 mg/dL    | mg/dL |

### 6.4.1 Data Model

| Field          | Type   | Required | Description                                                                                                                                                                                                                                       |
|----------------|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| encounter_code | string | YES      | The code of the patient encounter when the vital signs were recorded.<br>It <b>can match</b> different encounter types depending on the context: <ul style="list-style-type: none"> <li><b>Triage.code</b> – if recorded during triage</li> </ul> |

|              |                  |     |                                                                                                                                                                                                                                                                          |
|--------------|------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                  |     | <ul style="list-style-type: none"> <li>• <b>IPD.code</b> – if recorded during an inpatient visit</li> <li>• <b>Emergency.code</b> – if recorded during an emergency visit</li> <li>• <b>ProgressNote.code</b> – if recorded during a routine daily assessment</li> </ul> |
| recorded_at  | datetime         | YES | Date and time vital signs were recorded                                                                                                                                                                                                                                  |
| recorded_by  | string           | YES | Name of healthcare provider who recorded the vital signs                                                                                                                                                                                                                 |
| title        | string           | YES | Doctor or nurse                                                                                                                                                                                                                                                          |
| observations | array of objects | YES | Array of vital sign measurements                                                                                                                                                                                                                                         |

### 6.4.2 Observation in Vital Sign

| Field | Type    | Required | Description            |
|-------|---------|----------|------------------------|
| name  | string  | YES      | Name of the vital sign |
| value | numeric | YES      | Measured value         |

### 6.4.3 Example

```
{
  "vital_signs": [
    {
      "encounter_code": "E417001",
      "recorded_at": "2025-03-17T09:40:00.000+07:00",
      "recorded_by": "Khun Srey Mom",
      "title": "Doctor",
      "observations": [
        {
          "name": "Systolic Blood Pressure",
          "value": 135,
        },
        {
          "name": "Diastolic Blood Pressure",
          "value": 85,
        },
        {
          "name": "Heart Rate",
          "value": 92,
        },
        {
          "name": "Respiratory Rate",
          "value": 18,
        },
        {
          "name": "Temperature",
          "value": 38.7,
        },
      ],
    },
  ],
}
```

```

    "name": "Blood Oxygen",
    "value": 97,
  },
  {
    "name": "Random Blood Glucose",
    "value": 110,
  }
]
}
]
}

```

## 6.5 MEDICAL HISTORIES

Medical History records the patient's past and current health conditions for clinical reference.

### 6.5.1 Components

| Category           | Section                         | Description                                                                                                    |
|--------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------|
| Current Illness    | History of Present Illness      | Detailed narrative of symptoms — onset, duration, severity, aggravating/relieving factors, associated symptoms |
| Current Illness    | Current Medication              | Current medications (name, dose, frequency), over-the-counter drugs, supplements                               |
| Health History     | Past Medical History            | Previous diagnoses (e.g., diabetes, hypertension), hospitalizations, chronic conditions                        |
| Health History     | Past Surgical History           | Details of past surgeries (type, date, complications, if any)                                                  |
| Health History     | Allergies                       | Known food, drug, or environmental reaction                                                                    |
| Health History     | Immunization                    | Vaccination status — especially for tetanus, flu, COVID-19, and childhood vaccines                             |
| Women's Health     | Obstetric & Gynecologic History | For female patients: menarche, menstruation pattern, pregnancies, contraception, and menopause                 |
| Family & Lifestyle | Family History                  | Hereditary conditions in parents, siblings, and grandparents. Example: heart disease, cancer                   |
| Family & Lifestyle | Social History                  | Lifestyle factors: tobacco use, alcohol, drug use, occupation, living situation, marital status                |

**Note:** the medical history detailed information must use these section names.

### 6.5.2 Data Model

| Field           | Type                        | Required | Description                    |
|-----------------|-----------------------------|----------|--------------------------------|
| medical_history | array of conditions objects | NO       | Array of conditions (nullable) |

### 6.5.3 Condition Model

| Field          | Type            | Required | Description                               |
|----------------|-----------------|----------|-------------------------------------------|
| patient_code   | string          | YES      | Reference to the patient unique code      |
| visit_code     | string          | YES      | Reference to visit the unique code        |
| encounter_code | text            | YES      | Reference to encounter a unique code      |
| name           | string          | YES      | Name of the condition                     |
| value          | array of string | YES      | Array of value in string of the condition |

### 6.5.4 Example

```
{
  "medical_histories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Immunization History",
      "value": [
        "BCG",
        "OPV",
        "DTP",
        "Measles",
        "COVID-19"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Allergies",
      "value": [
        "Prahok",
        "Shrimp paste",
        "Penicillin"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Past Surgical History",
      "value": [
        "Appendectomy Surgery"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Past Medical History",

```

```

    "value": [
      "Dengue fever (2020)",
      "Typhoid (2018)",
      "Cesarean section at Calmette Hospital (2019)"
    ]
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Family History",
    "value": [
      "Mother with diabetes",
      "Father with hypertension and stroke at age 60"
    ]
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Current Illness History",
    "value": [
      "Patient reports high fever with chills occurring every 48
hours, accompanied by headache and joint pain. Symptoms started 3 days
ago after returning from a trip to Mondulkiri province. No prior history
of malaria. No medication taken yet."
    ]
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Current Medication",
    "value": [
      "Metformin 500mg",
      "Amlodipine 5mg"
    ]
  }
]
}

```

\*Note: if the value is empty or not available, the object must not be included

## 6.6 PHYSICAL EXAMINATIONS

Physical Examination contains clinical findings from examination of a patient.

### 6.6.1 Assessment Points

| System/Section     | Assessment Points                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| General Appearance | Consciousness (alert/drowsy), Distress, Nutrition, Posture, Gait, Hygiene, Pallor, Jaundice, Cyanosis, Edema |

|                        |                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------|
| Glasgow Coma Scale     | Eye opening, Verbal response, Motor response                                                                   |
| Head                   | Scalp condition, Hair distribution, Lesions, Deformities                                                       |
| Eyes                   | Pupils (size, reaction), Conjunctiva (pallor), Sclera (icterus), Eye movements                                 |
| Ears                   | Canal and Tympanic membrane, Discharge, Hearing                                                                |
| Nose                   | Nasal mucosa, Septum deviation, Discharge, Polyps                                                              |
| Throat/Mouth           | Tonsils, Pharynx, Uvula, Oral mucosa, Tongue, Teeth                                                            |
| HEENT*                 | Information of Head, Eyes, Ears, Nose, Throat can be combined                                                  |
| Neck                   | Lymph nodes, Thyroid gland, Tracheal position, JVP                                                             |
| Cardiovascular System  | Inspection (chest), Palpation (apex beat), Auscultation (S1/S2, murmurs), Peripheral pulses                    |
| Respiratory System     | Chest symmetry, Expansion, Percussion (dullness), Breath sounds, Crepitations, Wheezing                        |
| Digestive System       | Inspection (distension), Palpation (tenderness, organ size), Percussion, Bowel sounds                          |
| Genitourinary System   | External genitalia (if applicable), Bladder palpation, Costovertebral angle tenderness, Discharge (if present) |
| Musculoskeletal System | Joint swelling, Deformity, Range of motion, Muscle tone/strength, Spine, Gait                                  |
| Nervous System         | Mental status, Cranial nerves, Motor/sensory function, Reflexes, Coordination                                  |
| Skin                   | Rashes, Lesions, Color changes, Ulcers, Turgor, Temperature                                                    |
| Lymph Nodes            | Cervical, Axillary, Inguinal – enlargement, tenderness, mobility                                               |

### 6.6.2 Data Model

| Field                 | Type            | Required | Description                           |
|-----------------------|-----------------|----------|---------------------------------------|
| physical_examinations | array of object | NO       | Array of assessment points (nullable) |

### 6.6.3 Physical Examination Model

| Field          | Type    | Required | Description                            |
|----------------|---------|----------|----------------------------------------|
| patient_code   | string  | YES      | Reference to patient unique code       |
| visit_code     | string  | YES      | Reference to visit unique code         |
| encounter_code | string  |          | Reference to encounter unique code     |
| name           | string  | YES      | Name for the physical assessment point |
| value          | various | YES      | Value of the assessment result         |
| value_type     | string  | YES      | Type of Value                          |
| value_unit     | string  |          | Measurement unit represent the value   |

### 6.6.4 Example

```
{
  "physical_examinations": [
    {
```

```

    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "General Appearance",
    "value": "Ill-appearing with intermittent shivering",
    "value_type": "text",
    "value_unit": null
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E417001",
    "name": "Skin",
    "value": "Warm, jaundiced, no rash",
    "value_type": "text",
    "value_unit": null
  },
  {
    "patient_code": "P25003877",
    "visit_code": "V420650",
    "encounter_code": "E317001",
    "name": "Digestive System",
    "value": [
      {
        "inspection": "Flat, no visible masses"
      },
      {
        "palpation": "Soft, mild RUQ tenderness, spleen palpable 1cm
below costal margin"
      },
      {
        "percussion": "No shifting dullness"
      },
      {
        "auscultation": "Normal bowel sounds"
      }
    ],
    "value_type": "complex",
    "value_unit": null
  }
]
}

```

## 6.7 LABORATORIES

### 6.7.1 Data Model

| Field        | Type            | Required | Description                          |
|--------------|-----------------|----------|--------------------------------------|
| laboratories | array of object | NO       | Array of laboratory tests (nullable) |

### 6.7.2 Laboratories Model

| Field          | Type            | Required | Description                                 |
|----------------|-----------------|----------|---------------------------------------------|
| patient_code   | string          | YES      | Reference to patient unique code            |
| visit_code     | string          | YES      | Reference to visit unique code              |
| encounter_code | string          |          | Reference to encounter unique code          |
| srequest_code  | string          | YES      | Unique code represents the lab request      |
| requested_at   | datetime        | YES      | Date and time laboratory were requested     |
| requested_by   | string          | YES      | Name of doctor who requested the laboratory |
| title          | string          |          | Name of doctor                              |
| results        | array of object | YES      | Array of laboratory objects                 |

### 6.7.3 Result Model

| Field           | Type     | Required | Description                                                         |
|-----------------|----------|----------|---------------------------------------------------------------------|
| name            | string   | YES      | The item of lab result                                              |
| category        | string   | YES      | Category that lab result fall under (e.g. biochemistry, hematology) |
| value           | various  | YES      | Measured value                                                      |
| value_type      | string   | YES      | Type of value                                                       |
| value_unit      | string   | YES      | Unit of measurement                                                 |
| reference_range | string   | YES      | Normal range for the measurement                                    |
| interpretation  | string   | YES      | Interpretation of the result                                        |
| verified_at     | datetime | YES      | Date and time of doctor who verified                                |
| verified_by     | string   | YES      | Name of healthcare provider who verified                            |
| recorded_at     | datetime | YES      | Date and time of doctor who recorded                                |
| recorded_by     | string   | YESs     | Name of healthcare provider who recorded                            |

### 6.7.4 Example – Structured

```
{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-03-17T10:30:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "title": "Doctor",
      "results": [
        {
          "name": "Malaria Blood Smear",
          "category": "Parasitology",
          "value": "Positive for P. falciparum",
          "value_type": "string",
```

```

        "value_unit": null,
        "reference_range": "Negative",
        "interpretation": "Positive",
        "verified_at": "2025-03-17T10:50:00.000+07:00",
        "verified_by": "Dr. Meas Chenda",
        "recorded_at": "2025-03-17T10:52:00.000+07:00",
        "recorded_by": "Meas Chandaravuth"
    },
    {
        "name": "Parasitemia",
        "category": "Parasitology",
        "value": 2.5,
        "value_type": "float",
        "value_unit": "%",
        "reference_range": "0",
        "interpretation": "High",
        "verified_at": "2025-03-17T10:50:00.000+07:00",
        "verified_by": "Dr. Meas Chenda",
        "recorded_at": "2025-03-17T10:52:00.000+07:00",
        "recorded_by": "Meas Chandaravuth"
    },
    {
        "name": "Hemoglobin",
        "category": "Hematology",
        "value": 11.2,
        "value_type": "float",
        "value_unit": "g/dL",
        "reference_range": "13.5-17.5",
        "interpretation": "Low",
        "verified_at": "2025-03-17T10:50:00.000+07:00",
        "verified_by": "Dr. Meas Chenda",
        "recorded_at": "2025-03-17T10:52:00.000+07:00",
        "recorded_by": "Meas Chandaravuth"
    }
]
}

```

### 6.7.5 Example - Complex

```

{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-04-22T09:15:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "results": [
        {
          "name": "Complete Blood Count",

```

```

"category": "Hematology",
"value": {
  "Hemoglobin": {
    "result": "11.2 g/dL",
    "reference": "13.5-17.5 g/dL",
    "interpretation": "Low"
  },
  "White Blood Cells": {
    "result": "4.5 x10^3/μL",
    "reference": "4.5-11.0 x10^3/μL",
    "interpretation": "Normal"
  },
  "Red Blood Cells": {
    "result": "4.1 x10^6/μL",
    "reference": "4.5-5.9 x10^6/μL",
    "interpretation": "Low"
  },
  "Platelets": {
    "result": "120 x10^3/μL",
    "reference": "150-450 x10^3/μL",
    "interpretation": "Low"
  },
  "Hematocrit": {
    "result": "34%",
    "reference": "41-50%",
    "interpretation": "Low"
  },
  "MCV": {
    "result": "83 fL",
    "reference": "80-96 fL",
    "interpretation": "Normal"
  },
  "MCH": {
    "result": "27.3 pg",
    "reference": "27-33 pg",
    "interpretation": "Normal"
  },
  "MCHC": {
    "result": "32.9 g/dL",
    "reference": "32-36 g/dL",
    "interpretation": "Normal"
  }
},
"value_type": "object",
"value_unit": null,
"reference_range": null,
"interpretation": null,
"verified_at": "2025-03-17T10:50:00.000+07:00",
"verified_by": "Dr. Meas Chenda",
"recorded_at": "2025-03-17T10:52:00.000+07:00",
"recorded_by": "Meas Chandaravuth"
},

```

```

{
  "name": "Liver Function",
  "category": "Chemistry",
  "value": {
    "ALT": {
      "result": "45 U/L",
      "reference": "7-56 U/L",
      "interpretation": "Normal"
    },
    "AST": {
      "result": "42 U/L",
      "reference": "10-40 U/L",
      "interpretation": "High"
    },
    "ALP": {
      "result": "110 U/L",
      "reference": "44-147 U/L",
      "interpretation": "Normal"
    },
    "Total Bilirubin": {
      "result": "1.2 mg/dL",
      "reference": "0.1-1.2 mg/dL",
      "interpretation": "High-normal"
    },
    "Direct Bilirubin": {
      "result": "0.3 mg/dL",
      "reference": "0.0-0.3 mg/dL",
      "interpretation": "High-normal"
    },
    "Total Protein": {
      "result": "7.2 g/dL",
      "reference": "6.0-8.3 g/dL",
      "interpretation": "Normal"
    },
    "Albumin": {
      "result": "4.0 g/dL",
      "reference": "3.4-5.4 g/dL",
      "interpretation": "Normal"
    }
  },
  "value_type": "object",
  "value_unit": null,
  "reference_range": null,
  "interpretation": null,
  "verified_at": "2025-03-17T10:50:00.000+07:00",
  "verified_by": "Dr. Meas Chenda",
  "recorded_at": "2025-03-17T10:52:00.000+07:00",
  "recorded_by": "Meas Chandaravuth"
}
]
}

```

```
]
}
```

### 6.7.6 Example – File

```
{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-04-22T11:00:00.000+07:00",
      "requested_by": "Dr. Tep Veasna",
      "results": [
        {
          "name": "Blood Smear",
          "category": "Parasitology",
          "value":
["https://example.com/lab/EN20250422002/blood_smear_field1.jpg","https://example.com/lab/EN20250422002/blood_smear_field2.jpg","https://example.com/lab/EN20250422002/blood_smear_field3.jpg"]
          ,
          "value_type": "array",
          "value_unit": null,
          "reference_range": null,
          "interpretation": "Positive for P. falciparum",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        },
        {
          "name": "Microbiology Culture",
          "category": "Microbiology",
          "value":
["https://example.com/lab/EN20250422002/culture_plate1.jpg","https://example.com/lab/EN20250422002/culture_plate2.jpg","https://example.com/lab/EN20250422002/gram_stain.jpg","https://example.com/lab/EN20250422002/sensitivity_test.jpg"],
          "value_type": "array",
          "value_unit": null,
          "reference_range": null,
          "interpretation": "Positive for Salmonella typhi",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        }
      ]
    }
  ]
}
```

```
]
}
```

## 6.8 IMAGERIES

### 6.8.1 Data Model

| Field     | Type            | Required | Description                                                         |
|-----------|-----------------|----------|---------------------------------------------------------------------|
| imageries | array of object | NO       | Array of medical imaging records for a patient encounter (nullable) |

### 6.8.2 Imageries Model

| Field          | Type            | Required | Description                                               |
|----------------|-----------------|----------|-----------------------------------------------------------|
| patient_code   | string          | YES      | Reference to patient's unique code                        |
| visit_code     | string          | YES      | Reference to visit the unique code                        |
| encounter_code | string          |          | Reference to encounter a unique code                      |
| request_code   | string          | YES      | Unique code represents the lab request                    |
| requested_at   | datetime        | YES      | Date and time of the imagery requested                    |
| requested_by   | string          | YES      | Name of the healthcare provider who requested the imagery |
| title          | string          | YES      | Name of doctor                                            |
| results        | array of object | YES      | Array of imagery objects                                  |

### 6.8.3 Result Model

| Field       | Type            | Required | Description                                              |
|-------------|-----------------|----------|----------------------------------------------------------|
| name        | string          | YES      | The item of the imaging result                           |
| category    | string          | YES      | Category that imaging falls under (e.g. radiology, ... ) |
| images      | array of string | YES      | Measured value                                           |
| result      | string          | YES      | Full result of the imagery                               |
| conclusion  | string          | YES      | Summary of the result                                    |
| verified_at | datetime        | YES      | Date and time that the sample was verified               |
| verified_by | string          | YES      | Name of healthcare provider who verified the sample      |
| recorded_at | datetime        | YES      | Date and time results were recorded                      |
| recorded_by | String          | YES      | Name of the healthcare provider who recorded the result  |

### 6.8.4 Example

```
{
  "imageries": [
    {
      "patient_code": "P25003877",
```

```

"visit_code": "V420650",
"encounter_code": "E417001",
"request_code": "IM2301",
"requested_at": "2025-03-17T11:30:00.000+07:00",
"requested_by": "Dr. Chhun Sovannarith",
"title": "title",
"results": [
  {
    "name": "Brain Scan",
    "category": "CT Scan",
    "images": [
      "img.com/result2.jpg"
    ],
    "result": "string",
    "conclusion": null,
    "verified_at": "2025-03-17T10:50:00.000+07:00",
    "verified_by": "Dr. Meas Chenda",
    "recorded_at": "2025-03-17T10:52:00.000+07:00",
    "recorded_by": "Meas Chandaravuth"
  }
],
},
{
  "patient_code": "P25003877",
  "visit_code": "V420650",
  "encounter_code": "E417001",
  "requested_at": "2025-03-17T11:35:00.000+07:00",
  "requested_by": "Dr. Chhun Sovannarith",
  "title": "Doctor",
  "results": [
    {
      "name": "Adomen Ultrasound",
      "category": "Ultrasound",
      "images": [
        "img.com/result1.jpg"
      ],
      "result": "string",
      "conclusion": null,
      "verified_at": "2025-03-17T10:50:00.000+07:00",
      "verified_by": "Dr. Meas Chenda",
      "recorded_at": "2025-03-17T10:52:00.000+07:00",
      "recorded_by": "Meas Chandaravuth"
    }
  ]
}
]
}

```

## 6.9 DIAGNOSIS

### 6.9.1 Diagnosis Types

| Diagnosis Type | Description                                                           |
|----------------|-----------------------------------------------------------------------|
| Differential   | A list of possible conditions that could explain the symptoms         |
| In             | The condition is suspected when the patient is first admitted         |
| Out            | Final diagnosis after treatment and evaluation, recorded at discharge |
| Primary        | The main reason for the current hospital visit or treatment           |
| Secondary      | Coexisting conditions that influence care or outcomes                 |

### 6.9.2 Diagnosis Model

| Field                 | Type     | Require | Description                                           |
|-----------------------|----------|---------|-------------------------------------------------------|
| patient_code          | string   | YES     | Reference to patient unique code                      |
| visit_code            | string   | YES     | Reference to visit unique code                        |
| encounter_code        | string   |         | Reference to encounter unique code                    |
| diagnosis_type        | string   | YES     | Name of diagnosis type as above table                 |
| diagnosis_code        | string   | YES     | ICD-10 code for the diagnosis                         |
| diagnosis_name        | string   | YES     | Name of the diagnosis                                 |
| diagnosis_description | string   |         | Detailed description of the diagnosis                 |
| diagnosed_at          | datetime | YES     | Date and time of diagnosis                            |
| diagnosed_by          | string   | YES     | Name of healthcare provider who made the diagnosis    |
| title                 | string   |         | Position of the healthcare provider (doctor or nurse) |

### 6.9.3 Example

```
{
  "diagnosis": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E317001",
      "diagnosis_type": "Primary",
      "diagnosis_code": "B50",
      "diagnosis_name": "Plasmodium falciparum malaria",
      "diagnosis_description": "Plasmodium falciparum malaria with cerebral complications",
      "diagnosed_at": "2025-03-17T10:05:00.000+07:00",
      "diagnosed_by": "Dr. Chhun Sovannarith",
      "title": "Doctor"
    }
  ]
}
```

## 6.10 SOAPS

SOAPS are structured clinical notes that record Subjective, Objective, Assessment, and Plan information. They may supplement detailed clinical data or provide a summary of the patient's clinical information

### 6.10.1 Data Model

| Field | Type            | Require | Description                                                                              |
|-------|-----------------|---------|------------------------------------------------------------------------------------------|
| soaps | array of object | NO      | Array of clinical notes written using the SOAP format for a patient encounter (nullable) |

### 6.10.2 SOAPS Model

| Field          | Type     | Require | Description                                               |
|----------------|----------|---------|-----------------------------------------------------------|
| encounter_code | string   | YES     | Reference to encounter unique code                        |
| subjective     | text     |         | Patient's reported symptoms and feelings                  |
| objective      | text     |         | Provider's objective                                      |
| assessment     | text     |         | Provider's assessment of the patient situation            |
| evaluation     | text     |         | Provider's summary or evaluation regarding the assessment |
| plan           | text     |         | Plan for treatment or next steps                          |
| code           | string   |         | Unique identifier for the SOAPS                           |
| recorded_at    | datetime |         | Date and time SOAPS was written                           |
| recorded_by    | string   |         | Name of healthcare provider who wrote the SOAPS           |
| title          | string   |         | Position of the healthcare provider (doctor or nurse)     |

### 6.10.3 Example

```
{
  "soaps": [
    {
      "encounter_code": "EN20250317001",
      "code": "SP3177998",
      "subjective": "Patient reports fever has decreased after first dose of medication. Still experiencing mild headache and fatigue. No chills since this morning. Able to tolerate oral intake well.",
      "objective": "Temperature 37.8°C (down from 38.7°C). Heart rate 85/min. Blood pressure 125/80 mmHg. Skin less jaundice. Abdomen remains soft with mild tenderness in RUQ.",
      "assessment": "Uncomplicated malaria showing initial response to treatment. Fever is trending down. No signs of complications.",
      "plan": "Continue artemether-lumefantrine as prescribed. Encourage oral hydration. Follow-up in clinic in 3 days. Return sooner if symptoms worsen or unable to tolerate medication.",
    }
  ]
}
```

```

    "recorded_at": "2026-02-27T03:56:57.000+07:00",
    "recorded_by": "Dr.Sochea",
    "title": "Doctor"
  }
]
}

```

## 6.11 PRESCRIPTIONS

A prescription contains all medications, dosages, and instructions for a patient during a particular encounter.

### 6.11.1 Data Model

| Field         | Type            | Require | Description                                                       |
|---------------|-----------------|---------|-------------------------------------------------------------------|
| prescriptions | array of object | NO      | Array of medications ordered by a doctor for a patient (nullable) |

### 6.11.2 Prescription Model

| Field          | Type             | Require | Description                                            |
|----------------|------------------|---------|--------------------------------------------------------|
| patient_code   | string           | YES     | Reference to the patient's unique code                 |
| visit_code     | string           | YES     | Reference to visit the unique code                     |
| encounter_code | string           |         | Reference to encounter a unique code                   |
| code           | string           | YES     | Unique identifier for the prescription                 |
| prescribed_at  | datetime         | YES     | Date and time the prescription was written             |
| prescribed_by  | string           | YES     | Name of healthcare provider who wrote the prescription |
| title          | string           |         | Position, whether doctor or nurse, or else             |
| medications    | array of objects | YES     | Array of medications is prescribed                     |

#### NOTE:

- ✎ **Medication dispenses during hospitalization:** Require encounter\_code linking to progress\_notes, outpatient, inpatient, surgery, or emergency
- ✎ **Medication dispensed for home use:** Require encounter\_code to be null

### 6.11.3 Medication Model

| Field         | Type   | Require | Description                          |
|---------------|--------|---------|--------------------------------------|
| medicine_code | string | YES     | Unique identifier for the medication |
| medicine_name | string | YES     | Name of the medicine                 |
| strength      | string |         | Strength of the medicine             |

|           |         |     |                                  |
|-----------|---------|-----|----------------------------------|
| method    | string  | YES | Method of administration         |
| unit      | string  | YES | Unit of measurement              |
| morning   | numeric |     | Morning dose                     |
| afternoon | numeric |     | Afternoon dose                   |
| evening   | numeric |     | Evening dose                     |
| night     | numeric |     | Night dose                       |
| days      | integer | YES | Duration of prescription in days |
| interval  | string  |     | Interval between doses           |
| note      | text    |     | Additional notes or instructions |

**NOTE:**

- ✎ If no **interval** => **morning, afternoon, evening, night** should be presented
- ✎ Strength is required if it is not presented in medicine\_name

#### 6.11.4 Dispense Model

| Field    | Type    | Require | Description                                               |
|----------|---------|---------|-----------------------------------------------------------|
| unit     | string  | YES     | Unit of measurement (e.g., "Boxes", "Bottles", "Tablets") |
| quantity | integer | YES     | Total amount dispensed to the patient                     |

**NOTE:**

- ✎ Dispense: Specifies the total quantity of the drug provided to the patient.
- ✎ If the dispense field is not provided, the system calculates the quantity using this fallback logic:
  1. **Quantity Formula:** (morning + afternoon + evening + night) × days
  2. **Zero Fallback:** If days is 0 or missing, the quantity defaults to 0
  3. **Unit:** Defaults to the unit specified in the usage instructions

**Example Calculation:**

1. Morning: 1, Afternoon: 0, Evening: 1, Night: 0
2. Days: 7
3. Calculated Quantity:  $(1 + 0 + 1 + 0) \times 7 = 14$  tablets

#### 6.11.5 Example

```
{
  "prescriptions": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "code": "RX20250317001",
      "prescribed_at": "2025-03-17T10:10:00.000+07:00",
      "prescribed_by": "Dr. Chhun Sovannarith",
    }
  ]
}
```

```

"title": "Doctor",
"medications": [
  {
    "medicine_code": "MED001",
    "medicine_name": "Artemether-Lumefantrine",
    "strength": "20/120mg",
    "form": "Tablet",
    "method": "Oral",
    "unit": "Tablet",
    "morning": 4,
    "afternoon": 0,
    "evening": 4,
    "night": 0,
    "days": 3,
    "interval": null,
    "note": "Take with food. Complete full course even if feeling better."
  },
  {
    "medicine_code": "MED002",
    "medicine_name": "Paracetamol",
    "strength": "500mg",
    "form": "Tablet",
    "method": "Oral",
    "unit": "Tablet",
    "morning": null,
    "afternoon": null,
    "evening": null,
    "night": null,
    "days": 3,
    "interval": "8 hours",
    "note": "Take as needed for fever or pain.",
    "dispense": {
      "quantity": 1,
      "unit": "Boxes"
    }
  }
]
}

```

## 6.12 REFERRAL

Referral contains information about patient transfers between healthcare facilities.

### 6.12.1 Data Model

| Field         | Type   | Description                              |
|---------------|--------|------------------------------------------|
| referred_from | object | Information about the referring facility |
| referred_to   | object | Information about the receiving facility |

### 6.12.2 Detail Model

| Field             | Type     | Require | Description                                          |
|-------------------|----------|---------|------------------------------------------------------|
| code              | string   | YES     | Unique identifier for the referral                   |
| referral_number   | string   | YES     | Unique reference number for the referral             |
| transportation    | string   | YES     | Mode of transportation                               |
| reason            | string   | YES     | Reason for referral                                  |
| has_called        | boolean  |         | Whether the receiving facility was called in advance |
| caretaker_name    | string   |         | Name of the person accompanying the patient          |
| caretaker_phone   | string   |         | Contact number for the caretaker                     |
| referred_by       | string   |         | Name of the referring provider                       |
| referred_by_phone | string   |         | Contact number for the referring provider            |
| referred_at       | datetime |         | Date and time of referral                            |
| received_by       | string   |         | Name of the receiving provider                       |
| received_by_phone | string   |         | Contact number for the receiving provider            |
| received_at       | datetime |         | Date and time the patient was received               |
| medications       | string   |         | Medications that the patient is taking               |
| health_facility   | object   |         | receiving facility/ referring facility               |

### 6.12.3 Health Facility Model

| Field | Type   | Require | Description              |
|-------|--------|---------|--------------------------|
| name  | string | YES     | Health facility 's name  |
| code  | string | YES     | Health facility 's code  |
| label | string |         | Health facility 's label |
| type  | string |         | Health facility 's type  |

### 6.12.4 Example

```
{
  "referrals": {
    "referred_from": null,
    "referred_to": {
      "code": "REF20250317001",
      "referral_number": "HCPP/REF/2025/317",
      "transportation": "Ambulance",
      "health_facility": {
        "name": "string", //required
        "code": "string", //required
        "label": "string", //optional
        "type": "HF level" //optional
      },
      "reason": "Severe malaria with signs of organ dysfunction
requiring ICU care",
    }
  }
}
```

```

    "has_called": true,
    "caretaker_name": "Sok Channary",
    "caretaker_phone": "012345678",
    "referred_by": "Dr. Chhun Sovannarith",
    "referred_by_phone": "0987654321",
    "referred_at": "2025-03-17T11:30:00.000+07:00",
    "received_by": "Dr. Pich Vothy",
    "received_by_phone": "0712345678",
    "received_at": "2025-03-17T12:15:00.000+07:00",
    "medications": "Artemether-Lumefantrine 20/120mg (1 dose given at
10:15), IV Artesunate 120mg initiated at 11:00"
  }
}
}

```

## 6.13 CONSULTATIONS

A clinical encounter where a patient interacts with a healthcare provider for assessment, advice, or diagnosis, *without necessarily receiving a prescription or procedure*.

| Field         | Type            | Require | Description                                                               |
|---------------|-----------------|---------|---------------------------------------------------------------------------|
| consultations | Array of String | NO      | assesses, advises, and guides a patient regarding their health condition. |

Example:

```

{
  "visits": [
    {
      "health_facility_code": "HEFA-00002S75",
      "patient_code": "PT005852",
      "code": "VIS012455",
      "admission_type": "Walk-in",
      "visit_type": "OPD",
      "admitted_at": "2025-11-28T21:46:00.000+07:00",
      "discharge_type": "Authorizes",
      "discharged_at": "2025-11-28T22:23:00.000+07:00",
      "consultations": ["Do exercise regularly", "Drink more water"],
      {...}
    }
  ]
}

```

## 6.14 INVOICES

Invoice records track payments for healthcare services and medications.

### 6.14.1 Data Model

| Field          | Type     | Require | Description                        |
|----------------|----------|---------|------------------------------------|
| patient_code   | string   | YES     | Reference to patient unique code   |
| visit_code     | string   | YES     | Reference to visit unique code     |
| encounter_code | string   |         | Reference to encounter unique code |
| code           | string   | YES     | Invoice identifier                 |
| payment_type   | string   | YES     | Type of payment                    |
| invoice_date   | datetime | YES     | Invoice date                       |
| total          | number   | YES     | Total amount                       |
| created_at     | datetime | YES     | Creation timestamp                 |
| updated_at     | datetime |         | Update timestamp                   |
| cashier        | string   | YES     | Name of cashier                    |

### 6.14.2 Service Model

| Field            | Type    | Require | Description                            |
|------------------|---------|---------|----------------------------------------|
| invoice_code     | string  | YES     | Invoice identifier                     |
| service_code     | string  | YES     | Code identify the provided service     |
| service_name     | string  | YES     | Name of the service provided           |
| service_category | string  | YES     | Category name of the service           |
| price            | number  | YES     | Unit price                             |
| quantity         | number  |         | Number of service provided             |
| payment          | number  |         | Payment received from patient          |
| paid             | boolean |         | Payment status                         |
| discount_type    | string  |         | String whether in percentage or amount |
| discount         | number  |         | Discount amount or percentage          |

### 6.14.3 Medication Model

| Field         | Type    | Require | Description                          |
|---------------|---------|---------|--------------------------------------|
| invoice_code  | string  | YES     | Invoice identifier                   |
| medicine_code | string  | YES     | Code identifies the provided service |
| medicine_name | string  | YES     | Name of the service provided         |
| quantity      | number  | YES     | Number of medicines provided         |
| price         | number  | YES     | Unit price                           |
| payment       | number  |         | Payment received from patient        |
| paid          | boolean |         | Payment status                       |

|               |        |  |                                        |
|---------------|--------|--|----------------------------------------|
| discount_type | string |  | String whether in percentage or amount |
| discount      | number |  | Discount amount or percentage          |

#### 6.14.4 Example

```
{
  "invoices": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "code": "INV20250317001",
      "payment_type": "HEF",
      "invoice_date": "2025-03-17T11:00:00.000+07:00",
      "total": 120000,
      "created_at": "2025-03-17T11:00:00.000+07:00",
      "updated_at": "2025-03-17T11:00:00.000+07:00",
      "cashier": "Sok Kunthea",
      "services": [
        {
          "invoice_code": "INV20250317001",
          "service_code": "SRV001",
          "service_name": "Blood Smear for Malaria",
          "service_category": "Lab Test",
          "quantity": 1,
          "price": 15000,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        }
      ],
      "medications": [
        {
          "invoice_code": "INV20250317001",
          "medicine_code": "MED001",
          "medicine_name": "Artemether-Lumefantrine 20/120mg",
          "quantity": 24,
          "unit": "Tablet",
          "price": 1500,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        },
        {
          "invoice_code": "INV20250317001",
          "medicine_code": "MED002",
          "medicine_name": "Paracetamol 500mg",
          "quantity": 24,
          "unit": "Tablet",
          "price": 1000,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        }
      ]
    }
  ]
}
```

```

        "paid": false,
        "discount_type": null,
        "discount": 0
    }
]
}
]
}

```

## 7 SPECIFIC ENCOUNTERS

---

### 7.1 INPATIENT ADMISSION

Inpatient Admission represents a patient's hospital stay where at least one overnight stay is required for treatment, observation, or procedures.

#### 7.1.1 Data Model

| Field          | Type            | Require | Description                                                 |
|----------------|-----------------|---------|-------------------------------------------------------------|
| visit_code     | string          | YES     | Reference to the visit unique code                          |
| is_new         | boolean         |         | Field to distinguish new cases from follow-ups              |
| code           | string          | YES     | Unique identifier for the operation                         |
| symptoms       | Array of string |         | Primary symptom or problem reported by the doctor(nullable) |
| ward           | string          | YES     | Name of the ward or clinic, or sub-clinic                   |
| service_type   | string          | YES     | Name of the service or purpose of inpatient (nullable)      |
| started_at     | datetime        | YES     | Date and time operation started                             |
| ended_at       | datetime        |         | Date and time the operation ended                           |
| encountered_by | string          | YES     | Name of the healthcare provider                             |
| title          | string          |         | Position, whether doctor or nurse, or else                  |
| created_at     | datetime        | YES     | Record creation timestamp                                   |
| updated_at     | datetime        | YES     | Record the last update timestamp                            |

#### 7.1.2 Example

```

{
  "inpatients": [
    {
      "visit_code": "V420650",
      "is_new": false,
      "code": "E317001",

```

```

    "symptoms": ["fever", "red skin"],
    "ward": "General IPD",
    "service_type": "Internal Medicine",
    "started_at": "2025-03-20T14:00:00.000+07:00",
    "ended_at": "2025-03-23T10:00:00.000+07:00",
    "encountered_by": "Dr. Chhun Sovannarith",
    "title": "Doctor",
    "created_at": "2025-03-20T14:00:00.000+07:00",
    "updated_at": "2025-03-23T10:00:00.000+07:00"
  }
]
}

```

## 7.2 OUTPATIENT

Outpatient represents a patient visit for treatment or consultation without requiring an overnight hospital stay.

### 7.2.1 Data Model

| Field          | Type            | Require | Description                                                 |
|----------------|-----------------|---------|-------------------------------------------------------------|
| visit_code     | string          | YES     | Reference to the visit unique code                          |
| is_new         | boolean         |         | Field to distinguish new cases from follow-ups              |
| code           | string          | YES     | Unique identifier for the operation                         |
| symptoms       | Array of string |         | Primary symptom or problem reported by the doctor(nullable) |
| ward           | string          | YES     | Name of the ward or clinic, or sub-clinic                   |
| service_type   | string          | YES     | Name of the service or purpose of outpatient (nullable)     |
| started_at     | datetime        | YES     | Date and time the operation started                         |
| ended_at       | datetime        |         | Date and time the operation ended                           |
| encountered_by | string          | YES     | Name of the healthcare provider                             |
| title          | string          |         | Position, whether doctor or nurse, or else                  |
| created_at     | datetime        | YES     | Record creation timestamp                                   |
| updated_at     | datetime        | YES     | Record the last update timestamp                            |

### 7.2.2 Example

```

{
  "outpatients": [
    {

```

```

    "visit_code": "V420650",
    "code": "E317000",
    "symptoms": ["fever", "red skin"],
    "ward": "General Inpatient",
    "service_type": "General Medicine",
    "started_at": "2025-03-17T09:35:00.000+07:00",
    "ended_at": "2025-03-17T10:15:00.000+07:00",
    "encountered_by": "Dr. Chhun Sovannarith",
    "title": "Doctor",
    "created_at": "2025-03-20T14:00:00.000+07:00",
    "updated_at": "2025-03-23T10:00:00.000+07:00",
    "created_at": "2025-03-17T09:35:00.000+07:00",
    "updated_at": "2025-03-17T10:15:00.000+07:00"
  }
]
}

```

## 7.3 EMERGENCY

Emergency represents a patient visit requiring immediate attention for acute or potentially life-threatening conditions.

### 7.3.1 Data Model

| Field          | Type            | Required | Description                                                 |
|----------------|-----------------|----------|-------------------------------------------------------------|
| visit_code     | string          | YES      | Reference to the visit unique code                          |
| is_new         | boolean         |          | Field to distinguish new cases from follow-ups              |
| code           | string          | YES      | Unique identifier for the operation                         |
| symptoms       | Array of string |          | Primary symptom or problem reported by the doctor(nullable) |
| ward           | string          | YES      | Name of the ward or clinic, or sub-clinic                   |
| service_type   | string          | YES      | Name of the service type or purpose (nullable)              |
| started_at     | datetime        | YES      | Date and time the encounter started                         |
| ended_at       | datetime        |          | Date and time the encounter ended                           |
| emergency_note | text            | YES      | Detailed notes about the emergency (nullable)               |
| encountered_by | string          | YES      | Name of the healthcare provider                             |
| title          | string          |          | Position, whether doctor or nurse, or else                  |
| created_at     | datetime        | YES      | Record creation timestamp                                   |
| updated_at     | datetime        | YESs     | Record the last update timestamp                            |

### 7.3.2 Example

```
{
  "emergencies": [
    {
      "visit_code": "V420650",
      "code": "E317001",
      "is_new": false,
      "ward": "Emergency",
      "symptoms": ["None"],
      "service_type_name": "Emergency Medicine",
      "started_at": "2025-03-17T15:30:00.000+07:00",
      "ended_at": "2025-03-17T17:45:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00",
      "emergency_notes": "Patient brought by family after motorcycle
accident. GCS 14/15. No loss of consciousness. Skull X-ray negative for
fracture. CT not available. Multiple abrasions on right side. No
apparent fractures. Wound cleaning and dressing performed. Tetanus
prophylaxis given.",
      "created_at": "2025-03-17T15:30:00.000+07:00",
      "updated_at": "2025-03-17T17:45:00.000+07:00"
    }
  ]
}
```

## 7.4 SURGERY

Surgery refers to medical procedures performed on a patient in an operating theatre to treat, repair, or diagnose a condition, usually under anesthesia.

### 7.4.1 Data Model

| Field          | Type            | Required | Description                                                 |
|----------------|-----------------|----------|-------------------------------------------------------------|
| visit_code     | string          | YES      | Reference to the visit unique code                          |
| is_new         | boolean         |          | Field to distinguish new cases from follow-ups              |
| encounter_code | string          |          | Unique identifier for the encounter result in operation     |
| code           | string          | YES      | Unique identifier for the operation                         |
| symptoms       | Array of string |          | Primary symptom or problem reported by the doctor(nullable) |
| theater_name   | string          | YES      | Name of the operating theater                               |
| service_type   | string          | YES      | Name of the operation                                       |
| started_at     | datetime        | YES      | Date and time the operation started                         |

|                          |                  |     |                                                                                                                 |
|--------------------------|------------------|-----|-----------------------------------------------------------------------------------------------------------------|
| ended_at                 | datetime         | YES | Date and time the operation ended                                                                               |
| reason                   | string           | YES | Reason for operating                                                                                            |
| anesthesia_type          | string           |     | Type of anesthesia used                                                                                         |
| procedure_notes          | text             | YES | Detailed notes about the procedure                                                                              |
| complications            | array of strings |     | Any complications during the operation                                                                          |
| specimens                | array of strings |     | Specimens collected during the operation                                                                        |
| pre_operative_diagnosis  | string           |     | The clinical diagnosis made BEFORE surgery, based on symptoms, physical examination, and investigations.        |
| post_operative_diagnosis | string           |     | The final diagnosis confirmed AFTER the surgery, based on what the surgeon actually found during the operation. |
| blood_loss               | string           |     | Amount of blood loss during surgery (including unit)                                                            |
| surgeon_name             | string           | YES | Name of the surgeon                                                                                             |
| anesthetist_name         | string           |     | Name of the anesthetist                                                                                         |
| assistant_names          | array of strings |     | Names of assistants                                                                                             |
| created_at               | datetime         | YES | Record creation timestamp                                                                                       |
| updated_at               | datetime         |     | Record the last update timestamp                                                                                |
| details                  | object           |     | Details how the surgery was performed                                                                           |

### 7.4.2 Detail Model

| Field    | Type   | Required | Description                           |
|----------|--------|----------|---------------------------------------|
| position | string | YES      | How the patient lies during surgeries |

### 7.4.3 Example

```
{
  "surgeries": [{
    "visit_code": "V420650",
    "encounter_code": "E401001",
    "code": "SUR20250401001",
    "theater_name": "Operating Theater 2",
    "service_type": "Laparoscopic Cholecystectomy",
    "started_at": "2025-04-01T08:30:00.000+07:00",
    "ended_at": "2025-04-01T10:15:00.000+07:00",
    "reason": "Symptomatic gallstones with recurrent biliary colic",
    "anesthesia_type": "General Endotracheal",
```

```

    "procedure_notes": "Patient positioned supine. Pneumoperitoneum
established with Veress needle. Four ports were placed. Gallbladder
dissected from liver bed with hook cautery. Cystic duct and artery
identified, clipped and divided. Gallbladder removed through umbilical
port. Hemostasis confirmed. No bile leak. Ports removed under direct
visualization. Wounds closed with 3-0 Vicryl and 4-0 Monocryl. Patient
tolerated procedure well.",
    "complications": [
        "Mild bleeding from liver bed, controlled with electrocautery"
    ],
    "specimens": [
        "Gallbladder with multiple stones",
        "Cystic fluid sample"
    ],
    "pre_operative_diagnosis": "Ancienne Sign nail du fémur gauche avec
plaie infecté (3ans)",
    "post_operative_diagnosis": "Ancienne Sign nail du fémur gauche avec
plaie infecté (3ans)",
    "details": {
        "position": "Décubitus dorsale"
    },
    "blood_loss": "5cl",
    "surgeon_name": "Dr. Sok Vichet",
    "anesthetist_name": "Dr. Meas Sokunthea",
    "assistant_names": ["Dr. Chhan Rathana" , "Nurse Van Bopha"s],
    "created_at": "2025-04-01T08:30:00.000+07:00",
    "updated_at": "2025-04-01T10:20:00.000+07:00"
  }
}

```

## 7.5 PROGRESS NOTES

Progress Note documents a patient’s ongoing assessment, treatment, and response during their care. It often uses the SOAP format.

### 7.5.1 Data Model

| Field          | Type     | Require | Description                                   |
|----------------|----------|---------|-----------------------------------------------|
| visit_code     | string   | YES     | Reference to the visit unique code            |
| encounter_code | string   |         | Reference to the parent encounter code        |
| code           | string   | YES     | Unique identifier for the daily progress note |
| started_at     | datetime | YES     | Date and time the assessment started          |
| ended_at       | datetime |         | Date and time the assessment ended            |
| encountered_by | string   | YES     | Name of the healthcare provider               |
| title          | string   | YES     | Position, whether doctor or nurse, or else    |
| created_at     | datetime | YES     | Record creation timestamp                     |
| updated_at     | datetime | YES     | Record the last update timestamp              |

## 7.5.2 Example

```
{
  "progress_notes ": [
    {
      "visit_code": "VS20250317001",
      "encounter_code": "E417001",
      "code": "E317001",
      "started_at": "2025-03-17T16:00:00.000+07:00",
      "ended_at": "2025-03-17T16:15:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title ": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00"
    }
  ]
}
```

Subject to be changed by NPCA

# 8 SPECIFIC CLINICAL DATA FORM COMPONENTS

## 8.1 INTRODUCTION TO CLINICAL DATA FORM

Clinical Data Forms are designed to provide a standardized structure for collecting additional clinical information that cannot be adequately represented by the common clinical data components such as Patient, Visit, Triage, Vital Signs, Medical History, Diagnosis, Prescription, Referral, or Invoice.

A unique Form Code identifies each Clinical Data Form and contains a defined set of clinical attributes. The Form Code identifies the type of clinical form and determines how the submitted data should be processed and stored.

### 8.1.1 Form Structure

A Clinical Data Form must contain the following information:

| Field          | Type   | Require | Description                                                                                                                                                                                                           |
|----------------|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| visit_code     | String | YES     | Reference to the visit unique code                                                                                                                                                                                    |
| encounter_code | String | YES     | A unique encounter identifier that references the code of the related clinical encounter.<br>Its value must match exactly one of the following, depending on the encounter type:<br>outpatient.code or inpatient.code |
| form_code      | String | YES     | Code of the clinical form.                                                                                                                                                                                            |

|           |        |     |                                      |
|-----------|--------|-----|--------------------------------------|
| form_name | String | YES | Name of the clinical form.           |
| form_data | Object | YES | Clinical data collected by the form. |

### 8.1.2 Example

```
"forms": [
  {
    "visit_code": "VIS071315",
    "form_code": "ENTY-000016Z",
    "form_name": "OPD Contraception",
    "encounter_code": "MED062210",
    "form_data": {
      "birth_control_type": "ថ្នាំគ្រាប់",
      "birth_control_customer_type": "ធ្លាប់ប្រើ"
    }
  }
]
```

## 8.2 DELIVERY FORM

The Delivery Form is used to record clinical information related to the delivery of a baby, including the delivery date and time, mode of delivery, maternal condition, delivery complications, and newborn information.

The form should be submitted when a delivery occurs or when the delivery event is clinically documented in the EMR.

### 8.2.1 Data Form Identifications

| Name                             | Type     | Require | Unit    | Description                                           |
|----------------------------------|----------|---------|---------|-------------------------------------------------------|
| maternal_delivery_status         | String   |         |         | ស្ថានភាពស្ត្រីសម្រាលកូន                               |
| delivery_number_of_newborn       | Number   | YES     |         | ចំនួនទារកទើបនឹងកើត                                    |
| delivery_number_of_pregnant      | Number   | YES     |         | ចំនួនមានផ្ទៃពោះ                                       |
| delivery_number_of_delivery      | Number   | YES     |         | ចំនួនសម្រាលកូន                                        |
| delivery_number_of_miscarriage   | Number   | YES     |         | ចំនួនរលូតកូន                                          |
| delivery_number_of_abortion      | Number   | YES     |         | ចំនួនរំលូតកូន                                         |
| delivery_pregnancy_in_weeks      | Number   | YES     | សប្តាហ៍ | អាយុផ្ទៃពោះគិតជាសប្តាហ៍                               |
| delivery_at                      | Datetime | YES     |         | កាលបរិច្ឆេទសម្រាល                                     |
| delivery_type                    | String   | YES     |         | ប្រភេទសម្រាលកូន                                       |
| delivery_oxytocin_injection      | String   |         |         | ចាក់ Oxytocin ដល់ស្ត្រី(ក្នុងដំណាក់កាលទី៣នៃការសម្រាល) |
| delivery_oxytocin_injection_date | Datetime |         |         | កាលបរិច្ឆេទចាក់ Oxytocin                              |
| delivery_baby_born               | String   | YES     |         | ទារកកើត                                               |

|                                         |        |     |   |                             |
|-----------------------------------------|--------|-----|---|-----------------------------|
| delivery_baby_breathing_in_first_minute | String |     |   | ទារកដកដង្ហើមក្នុង១នាទីដំបូង |
| delivery_baby_sex                       | String | YES |   | ទារកភេទ                     |
| delivery_breastfeed_in_first_hour       | String |     |   | បំបៅដោះកូនក្នុង១ម៉ោងដំបូង   |
| delivery_baby_weight                    | Number | YES | g | ទម្ងន់ទារក (គិតជា g)        |

### 8.2.2 Validation Rule

| Rule               | Validation                              |
|--------------------|-----------------------------------------|
| Number of newborns | Must be $\geq 1$                        |
| Delivery Datetime  | Must not be later than the current date |
| Delivery Type      | Must use valid reference data           |
| Baby Born          | Must use valid reference data.          |
| Baby Sex           | Must use valid reference data.          |

## 8.3 OPD ANC FORM

The OPD ANC Form is used to record clinical information related to antenatal care provided to a pregnant woman during an outpatient visit.

The OPD ANC Form should be submitted when an outpatient encounter includes antenatal care (ANC).

### 8.3.1 Data Form Identifications

| Name                                 | Type   | Require | Unit    | Description                |
|--------------------------------------|--------|---------|---------|----------------------------|
| delivery_number_of_pregnant          | Number | YES     |         | ចំនួនមានផ្ទៃពោះ            |
| pregnancy_age                        | Number | YES     | សប្តាហ៍ | អាយុផ្ទៃពោះ (គិតជាសប្តាហ៍) |
| uterine_height_cm                    | Number |         | cm      | កម្ពស់ស្បូន (ស.ម.)         |
| fetal_heart_rate                     | Number |         | bpm     | ចង្វាក់បេះដូងកូន           |
| pregnancy_last_menstrual_period_date | Date   |         |         | ថ្ងៃមករដូវចុងក្រោយ         |

### 8.3.2 Validation Rule

| Rule                       | Validation                                                                                                                     |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Number of Pregnancies      | Must be $\geq 1$                                                                                                               |
| Pregnancy Age              | Integer/decimal, clinically valid range                                                                                        |
| Uterine Height             | Numeric, $\geq 0$ ; recommended to define a reasonable maximum                                                                 |
| Fetal Heart Rate           | Required when fetal heart rate is assessed; numeric; <b>range: 60–220 bpm</b>                                                  |
| Last Menstrual Period Date | Required if known; must be a valid date; <b>must not be in the future</b> ; should be <b>before or equal to the visit date</b> |

## 8.4 OPD PNC FORM

The OPD PNC (Postnatal Care) Form records postnatal visit information for both the mother and the newborn after delivery.

### 8.4.1 Data Form Identifications

| Name                                    | Type     | Require | Description               |
|-----------------------------------------|----------|---------|---------------------------|
| pnc_days_after_delivery                 | Number   | YES     | ចំនួនថ្ងៃក្រោយសម្រាលកូន   |
| delivery_at                             | Datetime | YES     | កាលបរិច្ឆេទសម្រាលកូន      |
| pnc_maternal_danger_signs_and_treatment | String   | YES     | សញ្ញាគ្រោះថ្នាក់របស់ម្តាយ |
| pnc_baby_danger_signs_and_treatment     | String   | YES     | សញ្ញាគ្រោះថ្នាក់របស់ទារក  |

### 8.4.2 Validation Rule

| Rule                | Validation                                                          |
|---------------------|---------------------------------------------------------------------|
| Days After Delivery | The value must be calculated from the delivery date and visit date. |
| Delivery Datetime   | Must not be later than the visit date                               |

## 8.5 ABORTION FORM

The Abortion Clinical Data Form records the patient's pregnancy/obstetric history, assessment at arrival, contraceptive service, and the abortion method provided during the encounter.

### 8.5.1 Data Form Identifications

| Name                                              | Type   | Require | Unit    | Description                                              |
|---------------------------------------------------|--------|---------|---------|----------------------------------------------------------|
| children_alive                                    | Number | YES     |         | ចំនួនកូនរស់                                              |
| delivery_number_of_misconriage                    | Number | YES     |         | ចំនួនលូតកូន                                              |
| delivery_number_of_abortion                       | Number | YES     |         | ចំនួនលូតកូន                                              |
| pregnancy_pregnancy_test                          | String |         |         | ការធ្វើតេស្តមានផ្ទៃពោះ                                   |
| problems_upon_arrival                             | String |         |         | អតិថិជនមានបញ្ហាទេ ពេលមកដល់                               |
| type_of_contraceptive_method_received             | String |         |         | តើអតិថិជនបានទទួលមធ្យោបាយពន្យារកំណើតប្រភេទណា ?            |
| safe_abortion_is_customer_get_method_before_leave | String | YES     |         | តើអតិថិជនបានទទួលមធ្យោបាយពន្យារកំណើតមុនចេញទៅផ្ទះដែលឬ ទេ ? |
| safe_abortion_uterine_size                        | Number | YES     | សប្តាហ៍ | ទំហំស្បូនគិតជាសប្តាហ៍                                    |
| safe_abortion_method                              | String | YES     |         | វិធីសាស្ត្រលូតកូន                                        |

### 8.5.2 Validation Rule

| Rule                                                    | Validation                                  |
|---------------------------------------------------------|---------------------------------------------|
| Children Alive                                          | Must be >= 0                                |
| Number of Miscarriages                                  | Must be >= 0                                |
| Number of Abortions                                     | Must be >= 0                                |
| Pregnancy Test                                          | Must use valid reference data               |
| Type of Contraceptive Method Received                   | Must use valid reference data               |
| Did the client receive birth control before going home? | Must use valid reference data               |
| Uterine size                                            | Must be >= 0, must be a valid numeric value |
| Abortion method                                         | Must use valid reference data               |

## 8.6 BIRTH CONTROL FORM

The Birth Control Form is a clinical data form used to record family planning and contraceptive services provided to a client. It captures the client's selected contraceptive method, counseling, provision of the method, and relevant follow-up information.

The form can be used during a family-planning visit or as part of postpartum/post-abortion care when contraception is provided.

### 8.6.1 Data Form Identifications

| Name                        | Type   | Require | Description                  |
|-----------------------------|--------|---------|------------------------------|
| birth_control_type          | String | YES     | ប្រភេទពន្យារកំណើត            |
| birth_control_customer_type | String | YES     | ប្រភេទពន្យារកំណើតរបស់អតិថិជន |

### 8.6.2 Validation Rule

| Rule                        | Validation                    |
|-----------------------------|-------------------------------|
| birth_control_type          | Must use valid reference data |
| birth_control_customer_type | Must use valid reference data |

## 8.7 CHILD PHYSICAL EXAMINATION (0-28 DAYS) FORM

The Child Physical Examination (0–28 Days) form is used to document the physical examination of a newborn from birth up to 28 days of age. It focuses on detecting congenital abnormalities, birth complications, infection, feeding problems, and other neonatal danger signs.

### 8.7.1 Data Form Identifications

| Name                    | Type   | Require | Unit | Description             |
|-------------------------|--------|---------|------|-------------------------|
| pnc_days_after_delivery | Number | YES     |      | ចំនួនថ្ងៃក្រោយសម្រាលកូន |

|                               |         |     |      |                                        |
|-------------------------------|---------|-----|------|----------------------------------------|
| child_growth_born_weight      | Number  | YES | g    | ទម្ងន់ពេលកើត (គិតជា g)                 |
| baby_maternal_fever_at_birth  | String  |     |      | ម្តាយក្តៅខ្លួនខ្លាំងនៅពេលសម្រាល        |
| baby_long_labor               | String  |     |      | ការសម្រាលមានរយៈពេលយូរ                  |
| term_or_preterm_birth         | Boolean | YES |      | កើតគ្រប់ខែ/មិនគ្រប់ខែ                  |
| baby_apgar_one_minute         | Number  |     | នាទី | APGAR (1 នាទី )                        |
| baby_apgar_ten_minutes        | Number  |     | នាទី | APGAR (10 នាទី )                       |
| baby_apgar_five_minutes       | Number  |     | នាទី | APGAR (5 នាទី )                        |
| baby_born_with_clubfoot       | String  |     |      | ទារកកើតបញ្ឆោតជើង                       |
| difficulty_with_breastfeeding | String  |     |      | បញ្ហាក្នុងការបៅ                        |
| baby_seizures                 | String  |     |      | ប្រកាច់ (ទារកប្រកាច់ )                 |
| meconium_passed_within24Hours | String  |     |      | គ្មានអាចម៍ព្រៃក្នុងកំឡុងពេល២៤ម៉ោងដំបូង |
| examination_results           | String  | YES |      | លទ្ធផលនៃការពិនិត្យ                     |
| other_observations            | String  |     |      | ការសង្កេតផ្សេងៗ( ការកម្សាយផ្សេងៗ )     |
| transfer                      | String  |     |      | បញ្ជូន                                 |

### 8.7.2 Validation Rule

| Rule                         | Validation                    |
|------------------------------|-------------------------------|
| Baby Maternal Fever at birth | Must use valid reference data |
| Baby long labor              | Must use valid reference data |
| Baby Premature birth         | Must use valid reference data |
| Baby Seizures                | Must use valid reference data |
| Baby Born with Clubfoot      | Must use valid reference data |

## 8.8 CHILD PHYSICAL EXAMINATION (1MONTH-5YEARS) FORM

Child Physical Examination (1 Month–5 Years) is a clinical form used by healthcare providers to document the physical examination findings of children from 1 month up to 5 years of age.

The purpose is to identify normal or abnormal physical findings, monitor the child's health and development, and support diagnosis and treatment.

### 8.8.1 Data Form Identifications

| Name                              | Type   | Require | Unit | Description                        |
|-----------------------------------|--------|---------|------|------------------------------------|
| child_had_serious_health_problems | String | YES     |      | កុមារធ្លាប់មានបញ្ហាសុខភាពធ្ងន់ធ្ងរ |

|                                |         |     |   |                                                     |
|--------------------------------|---------|-----|---|-----------------------------------------------------|
| child_growth_born_weight       | Number  | YES | g | ទម្ងន់ពេលកើត (គិតជា g)                              |
| term_or_preterm_birth          | Boolean | YES |   | កើតគ្រប់ខែ/មិនគ្រប់ខែ                               |
| complication_during_childbirth | Boolean |     |   | មានផលវិបាកពេលសម្រាល                                 |
| feeding_difficulties           | Boolean |     |   | បញ្ហាក្នុងការហូប/ផឹក                                |
| child_convulsions              | Boolean |     |   | ប្រកាច់                                             |
| toilet_training                | String  |     |   | ការចេះបន្ទោបង់( ធៀបនឹងកុមារផ្សេងដែលមានអាយុដូចគ្នា ) |
| difficulty_with_breastfeeding  | Boolean |     |   | បញ្ហាក្នុងការបៅ                                     |
| child_speech_development       | String  |     |   | ការនិយាយ                                            |
| social_play_development        | String  |     |   | ការលេង( ធៀបនឹងកុមារផ្សេងដែលមានអាយុដូចគ្នា )         |
| examination_results            | String  | YES |   | លទ្ធផលនៃការពិនិត្យ                                  |
| other_observations             | String  |     |   | ការសង្កេតផ្សេងៗ( ការកម្រោយផ្សេងៗ )                  |
| transfer                       | String  |     |   | បញ្ជូន                                              |

### 8.8.2 Validation Rule

| Rule              | Validation                    |
|-------------------|-------------------------------|
| Child Convulsions | Must use valid reference data |

## 9 STANDARD DATA REFERENCE GLOSSARY

### 9.1 PATIENT DEMOGRAPHIC

This section specifies the standardized patient information that should be recorded.

#### 9.1.1 Sex

| Code | Value | Description                        |
|------|-------|------------------------------------|
| F    | F     | Patient who is a female from birth |
| M    | M     | Patient who is a male from birth   |

#### 9.1.2 Marital Status

| Code        | Value     | Description                                                   |
|-------------|-----------|---------------------------------------------------------------|
| MAST-00M002 | Married   | Patient is legally married                                    |
| MAST-0000X1 | Single    | Patient has never been married                                |
| MAST-000S03 | Widowed   | Patient's spouse is deceased                                  |
| MAST-0000N4 | Divorced  | Patient has been legally divorced and has not remarried       |
| MAST-0000N5 | Separated | Patient is legally married but living apart from their spouse |

## 9.2 VITAL SIGNS OBSERVATIONS

This section defines the standard vital signs recorded during patient assessment.

| Concept                  | Description                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------|
| Temperature              | A float value representing the body's core thermal state.                                                        |
| Heart Rate               | An integer that functions as a real-time counter for the heart's pump cycle, measured in beats per minute (bpm). |
| Respiratory Rate         | An integer tracking the frequency of the body's primary resource-intake cycle (breathing).                       |
| Systolic Blood Pressure  | An integer that measures the peak pressure on the circulatory network during a pump cycle.                       |
| Diastolic Blood Pressure | An integer that measures the baseline pressure when the pump is in its resting phase.                            |
| Blood Oxygen             | A percentage integer that measures the saturation of the body's data carriers (hemoglobin) with oxygen.          |
| Random Blood Glucose     | An integer representing the amount of readily available fuel (glucose) in the system.                            |

## 9.3 VISIT & ADMINISTRATIVE DATA

This section covers categorical data related to patient visits, admissions, and billing.

### 9.3.1 Admission

| Code         | Value       | Description                                                    |
|--------------|-------------|----------------------------------------------------------------|
| ADTY-000J008 | Walk-in     | The patient comes to the facility without a prior appointment. |
| ADTY-000009S | Appointment | Patient arrives for a scheduled visit.                         |
| ADTY-0P0000A | Refer       | Patient is referred by another facility or provider.           |

### 9.3.2 Visit Types

| Code         | Value | Description                                             |
|--------------|-------|---------------------------------------------------------|
| VITY-00J000C | IPD   | Patient stays at the hospital for one or more nights.   |
| VITY-00L000D | OPD   | Patient receives care without being admitted overnight. |

### 9.3.3 Outcome Types

| Value          | Description                                                    |
|----------------|----------------------------------------------------------------|
| Unchanged      | No significant change in the patient's condition.              |
| Recovered      | Patient's condition has resolved.                              |
| Improved       | The condition has shown improvement but is not fully resolved. |
| Hopeless       | The patient's condition has no viable solution.                |
| Home treatment | The patient can continue to receive treatment at home.         |
| Deceased       | The patient passed away during the visit.                      |

#### 9.3.4 Discharge Types

| Value        | Description                                                            |
|--------------|------------------------------------------------------------------------|
| Authorized   | Patient is well enough to go home.                                     |
| Unauthorized | Patient chose to leave before formal discharge.                        |
| Deceased     | Patient passed away during the stay.                                   |
| Refer Out    | Patient is sent to another hospital or care facility.                  |
| Absconded    | Patient left without informing staff or completing a formal discharge. |

#### 9.3.5 Diagnosis Types

| Value        | Description                                                                   |
|--------------|-------------------------------------------------------------------------------|
| Differential | A list of possible conditions that could explain the symptoms.                |
| In           | The condition is suspected when the patient is first admitted.                |
| Out          | The final diagnosis after treatment and evaluation was recorded at discharge. |
| Primary      | The main reason for the current hospital visit or treatment.                  |
| Secondary    | Coexisting conditions that influence care or outcomes.                        |

#### 9.3.6 Payment Types

| Code         | Value                 | Description                                                          |
|--------------|-----------------------|----------------------------------------------------------------------|
| PATY-0000N09 | HEF (Poor)            | The patient is from a poor family, and payment is covered by the HEF |
| PATY-00002NF | HEF (Informal Worker) | Patient is an informal worker with payment covered by the HEF        |
| PATY-00003LD | HEF (At Risk)         | Patient is from a family at risk, and payment is covered by the HEF  |
| PATY-0000G3E | NSSF Employee         | Patient is an employee with payment is covered by the NSSF           |
| PATY-00B003F | NSSF Public Servant   | Patient is a public servant with payment is covered by the NSSF      |
| PATY-000T040 | NSSF Self-Employed    | Patient is self-employed, with payment is covered by the NSSF        |

#### 9.3.7 Card Types

| Value  | Description                                                            |
|--------|------------------------------------------------------------------------|
| HEF    | Health Equity Fund for Poor family                                     |
| HEFR   | Health Equity Fund for At Risk family                                  |
| HEFIY  | Health Equity Fund for Informal Worker with Yellow Card                |
| HEFIV  | Health Equity Fund for Informal Worker with Violet Card                |
| NSSF   | National Social Security Fund                                          |
| NSSP   | National Social Security Fund for Public servant                       |
| NSSPM  | National Social Security Fund for Public servant's family Member       |
| NSSE   | National Social Security Fund for Employees                            |
| NSSEM  | National social Security Fund for Employees' family Member             |
| NSSFS  | National Social Security Fund for Self-employed person                 |
| NSSFSM | National Social Security Fund for Self-employed person's family Member |

## 9.4 MEDICAL HISTORY COMPONENTS

This section outlines the key areas to be documented when taking a patient's medical history.

- **History of Present Illness:** A detailed narrative of symptoms—onset, duration, severity, aggravating/relieving factors, and associated symptoms.
- **Past Medical History:** Previous diagnoses (e.g., diabetes, hypertension), surgeries, hospitalizations, and chronic conditions.
- **Past Surgical History:** Details of past operations, including type, date, and any complications.
- **Current Medication:** A list of current medications (name, dose, frequency), over-the-counter drugs, and supplements.
- **Allergies:** Known drug, food, or environmental allergies and the type of reaction.
- **Family History:** Hereditary conditions in parents, siblings, and grandparents (e.g., heart disease, cancer).
- **Social History:** Lifestyle factors such as tobacco use, alcohol, drug use, occupation, living situation, and marital status.
- **Obstetric & Gynecologic History:** For female patients: menarche, menstruation pattern, pregnancies, contraception, and menopause.
- **Immunization History:** Vaccination status, especially for tetanus, flu, COVID-19, and common childhood vaccines.

## 9.5 PHYSICAL EXAMINATION COMPONENTS

This section details the systems to be assessed during a physical examination and what to record for each.

- **General Appearance:** Consciousness (alert/drowsy), Distress, Nutrition, Posture, Gait, Hygiene, Pallor, Jaundice, Cyanosis, Edema.
- **HEENT:** Head, Eyes, Ears, Nose, Throat/Mouth.
- **Neck:** Lymph nodes, Thyroid gland, Tracheal position, Jugular Venous Pressure (JVP).
- **Cardiovascular System:** Inspection (chest), Palpation (apex beat), Auscultation (S1/S2, murmurs), Peripheral pulses.
- **Respiratory System:** Chest symmetry, Expansion, Percussion (dullness), Breath sounds, Crepitations, Wheezing.
- **Abdomen:** Inspection (distension), Palpation (tenderness, organ size), Percussion, Bowel sounds.
- **Musculoskeletal System:** Joint swelling, Deformity, Range of motion, Muscle tone/strength, Spine, Gait.
- **Nervous System:** Mental status, Cranial nerves, Motor/sensory function, Reflexes, Coordination.
- **Skin:** Rashes, Lesions, Color changes, Ulcers, Turgor, Temperature.
- **Lymph Nodes:** Cervical, Axillary, Inguinal – note any enlargement, tenderness, or changes in mobility.
- **Glasgow Coma Scale:** Structured consciousness assessment using Eye, Verbal and Motor responses, producing a GCS score from 3–15.
- **Urogenital System:** Genitourinary examination: urinary/genital findings, tenderness, swelling, discharge, lesions, etc.
- **Mental Status:** Cognitive and behavioral assessment: appearance, behavior, mood, speech, orientation, memory, attention, thought process, etc.
- **Other body parts:** Examination findings involving body parts or anatomical areas not covered by the standard examination sections above.

## 9.6 FORM DATA

### 9.6.1 Birth Control Type

| Code         | Value               |
|--------------|---------------------|
| OBVA-S000099 | ថ្នាំគ្រាប់ស៊ីអូស៊ី |
| OBVA-00D0097 | ថ្នាំចាក់           |
| OBVA-00P0096 | កងដាក់ក្នុងស្បូន    |
| OBVA-00W009A | កងដាក់ក្រោមស្បែក    |
| OBVA-0000C98 | ស្រោមអនាម័យ         |
| OBVA-000159Z | ដោះកងដាក់ក្នុងស្បូន |
| OBVA-00A015B | ការដាក់កងក្នុងដៃ    |
| OBVA-000W15C | តាមដានកងដាក់ក្នុងដៃ |
| OBVA-C00015D | ដោះកងដៃចេញ          |
| OBVA-00015VE | ការផ្តល់ប្រឹក្សា    |

### 9.6.2 Birth Control Customer Type

| Code         | Value      |
|--------------|------------|
| OBVA-00001H1 | ធ្លាប់ប្រើ |
| OBVA-0000I10 | លើកដំបូង   |
| OBVA-0000I12 | ចាស់       |

### 9.6.3 Delivery Baby Born

| Code         | Value     |
|--------------|-----------|
| OBVA-00007ID | កើតរស់    |
| OBVA-00T007C | កើតស្លាប់ |

### 9.6.4 Delivery Baby Sex

| Code         | Value |
|--------------|-------|
| OBVA-00G007A | ប្រុស |
| OBVA-00007GB | ស្រី  |

### 9.6.5 Delivery Type

| Code         | Value        |
|--------------|--------------|
| OBVA-000B153 | សម្រាលធម្មតា |

|              |                     |
|--------------|---------------------|
| OBVA-00R0154 | សម្រាលដោយអន្តរាគមន៍ |
| OBVA-X000155 | សម្រាលដោយវះកាត់     |

#### 9.6.6 Safe Abortion Method

| Code         | Value                            |
|--------------|----------------------------------|
| OBVA-00H0156 | វិលូតដោយថ្នាំ                    |
| OBVA-000157X | វិលូតដោយបូមសុញ្ញាកាស             |
| OBVA-0001Z58 | វិលូតដោយពង្រីកមាត់ស្បូន និងយកចេញ |
| OBVA-000159Z | វិធីសាស្ត្រផ្សេងទៀត              |

#### 9.6.7 Type Of Contraceptive Method Received

| Code         | Value               |
|--------------|---------------------|
| OBVA-S000099 | ថ្នាំគ្រាប់ស៊ីអូស៊ី |
| OBVA-00D0097 | ថ្នាំចាក់           |
| OBVA-00P0096 | កងដាក់ក្នុងស្បូន    |
| OBVA-00W009A | កងដាក់ក្រោមស្បែក    |
| OBVA-0000C98 | ស្រោមអនាម័យ         |
| OBVA-0000I9B | បញ្ឈប់កំណើត         |

## 10 RELATED DOCUMENTS

- EMR Clinical Data Components
- OPD Sample JSON
- IPD Sample JSON
- Transfer OPD to IPD Sample JSON
- Referral Sample JSON
- Form Sample JSON